

Braided Operads

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1. Introduction

1 Introduction

Coming soon

2 Operads

Operads are a concept that originally occurred when studying loop spaces. Let ΩX be the space of based loops in a topological space X . Its structure reflects an interesting form of algebraic composition. The composition of loops is only associative up to homotopy equivalence meaning that loop spaces behave like a monoid in a homotopy-theoretic sense, but not in a strict algebraic sense. An operad can be thought of as a structure that encodes a family of operations, each of which has a certain number of inputs, and a way to compose these operations. The composition of operations in an operad satisfies associativity and unit laws.

Let $\alpha: [0, 1] \rightarrow X$ and $\beta: [0, 1] \rightarrow X$ be loops in X having the same base point x_0 . Their composition $\beta \circ \alpha$ is usually defined by splitting the unit interval in half, i.e.

$$\beta \circ \alpha(s) := \begin{cases} \alpha(2s) & \text{for } s \in [0, \frac{1}{2}] \\ \beta(2s - 1) & \text{for } s \in (\frac{1}{2}, 1]. \end{cases}$$

This concatenation in itself is not associative; consider $\alpha, \beta, \gamma \in \Omega X$. Then by the given definition $\gamma \circ (\beta \circ \alpha)$ and $(\gamma \circ \beta) \circ \alpha$ are different maps:

$$\gamma \circ (\beta \circ \alpha)(s) = \begin{cases} \alpha(4s) & \text{for } s \in [0, \frac{1}{4}] \\ \beta(4s - 1) & \text{for } s \in (\frac{1}{4}, \frac{1}{2}] \\ \gamma(2s - 1) & \text{for } s \in (\frac{1}{2}, 1] \end{cases}$$

$$(\gamma \circ \beta) \circ \alpha(s) = \begin{cases} \alpha(2s) & \text{for } s \in [0, \frac{1}{2}] \\ \beta(4s - 2) & \text{for } s \in (\frac{1}{2}, \frac{3}{4}] \\ \gamma(4s - 3) & \text{for } s \in (\frac{3}{4}, 1] \end{cases}$$

The choice of splitting the unit interval in half is merely a convention. Hence, composition of loops can be thought of as splitting intervals in an arbitrary manner; or in other words embedding unit intervals into a unit interval such that the image of the inner intervals are disjoint.

$$[[I_1] \dots [I_i] \dots [I_n]]$$

So, for an ordered family of loops $\{\alpha_i: I_i \rightarrow X\}_{1 \leq i \leq n}$ there is a unique map $f \in \text{Emb}(I_1, \dots, I_n, I)$ for each order preserving concatenation β such that $\beta \circ f(I_i) = \alpha_i$ for all $1 \leq i \leq n$. We denote the set of embeddings by $E(n) = \text{Emb}(I_1, \dots, I_n, I)$ for each $n \in \mathbb{N}$. For every $(f_k, (f_{j_i})_{1 \leq i \leq k}) \in E(k) \times E(j_1) \times \dots \times E(j_k)$ we obtain an embedding $f \in E(j_1 + \dots + j_k)$ by defining $f = f_k \circ (f_{j_1} \sqcup \dots \sqcup f_{j_k})$. This defines a composition map

$$\gamma: E(k) \times E(j_1) \times \dots \times E(j_k) \rightarrow E(j_1 + \dots + j_k).$$

Since the composition of maps is associative the following diagram commutes:

$$\begin{array}{ccc} E(k) \times \left(\prod_{s=1}^k E(j_s) \right) \times \left(\prod_{r=1}^j E(i_r) \right) & \xrightarrow{\gamma \times \text{Id}} & E(j) \times \left(\prod_{r=1}^j E(i_r) \right) \\ \downarrow s & & \downarrow \gamma \\ & & E(i) \\ & & \uparrow \gamma \\ E(k) \times \left(\prod_{s=1}^k \left(E(j_s) \times \left(\prod_{q=1}^{j_s} E(i_{g_{s-1}+q}) \right) \right) \right) & \xrightarrow{\text{Id} \times \gamma^s} & E(k) \times \left(\prod_{s=1}^k E(h_s) \right) \end{array}$$

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where s shuffles the cartesian product and

$$n, j_1, \dots, j_k, i_1, \dots, i_j \in \mathbb{N}$$

with

$$\sum_{1 \leq s \leq k} j_s = j, \sum_{1 \leq r \leq j} i_r = i, g_s = \sum_{0 \leq l \leq s} j_l \text{ and } h_s = \sum_{1 \leq q \leq j_s} i_{g_{s-1}+q}.$$

Further composing with the identity embedding in $E(1)$ does nothing. Thus, we have seen our first example of an operad, providing a framework in which the composition of loop spaces satisfies associativity.

In this way, operads provide a more general and flexible formalism for capturing the algebraic structures found in loop spaces and other topological constructs where composition occurs up to homotopy. Similarly to monoids that can be considered as single object categories, operads can be considered as single object multicategories. In this section, we will introduce the notion of a planar and a symmetric operad as they were introduced by May in [IM]. Then, we will move on to the more general notion of a colored operad following [JL]. We will establish a second equivalent definition of a colored operad that is purely categorical, meaning we can therefore also phrase it in an ∞ -categorical context. In the next section we will then consider braided operads and we work towards defining braided colored operads. Ultimately, our goal is a purely functorial definition of a braided colored operad such that we can lift the concept to an ∞ -categorical context as well.

2.1 Planar and Symmetric Operads

From now on, unless it is stated otherwise, let $\mathcal{C}$ be a symmetric monoidal category with tensor product $\otimes$ and tensor unit I . For an object C in $\mathcal{C}$, we write C^j for the j -th tensor power of C .

Definition 2.1. A *planar operad* $\mathcal{P}$ in $\mathcal{C}$ consists of a family of objects $(\mathcal{P}(j))_{j \in \mathbb{N}}$ together with a unit morphism $\eta: k \rightarrow \mathcal{P}(1)$ and a product morphism

$$\gamma: \mathcal{P}(k) \otimes \mathcal{P}(j_1) \otimes \dots \otimes \mathcal{P}(j_k) \rightarrow \mathcal{P}(j),$$

where $\sum_{1 \leq i \leq k} j_i = j$ such that the following associativity and unitality constraints hold:

- For any choice of integers

$$n, j_1, \dots, j_k, i_1, \dots, i_j$$

with

$$\sum_{1 \leq s \leq k} j_s = j, \sum_{1 \leq r \leq j} i_r = i, g_s = \sum_{0 \leq l \leq s} j_l \text{ and } h_s = \sum_{1 \leq q \leq j_s} i_{g_{s-1}+q}$$

the following diagram commutes:

$$\begin{array}{ccc} \mathcal{P}(k) \otimes \left(\bigotimes_{s=1}^k \mathcal{P}(j_s) \right) \otimes \left(\bigotimes_{r=1}^j \mathcal{P}(i_r) \right) & \xrightarrow{\gamma \otimes \text{Id}} & \mathcal{P}(j) \otimes \left(\bigotimes_{r=1}^j \mathcal{P}(i_r) \right) \\ \downarrow s & & \downarrow \gamma \\ & & \mathcal{P}(i) \\ & & \uparrow \gamma \\ \mathcal{P}(k) \otimes \left(\bigotimes_{s=1}^k \left(\mathcal{P}(j_s) \otimes \bigotimes_{q=1}^{j_s} \mathcal{P}(i_{g_{s-1}+q}) \right) \right) & \xrightarrow{\text{Id} \otimes (\bigotimes_s \gamma)} & \mathcal{P}(k) \otimes \left(\bigotimes_{s=1}^k \mathcal{P}(h_s) \right) \end{array}$$

where s shuffles the entries of the tensor product.

- For all $j \in \mathbb{N}$ the following diagrams commute:

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$$\begin{array}{ccc}
\mathcal{P}(j) \otimes k^j & \xrightarrow{\cong} & \mathcal{P}(j) \\
\text{Id} \otimes (\otimes_j \eta) \downarrow & \nearrow \gamma & \\
\mathcal{P}(j) \otimes \mathcal{P}(1)^j & &
\end{array}
\qquad
\begin{array}{ccc}
k \otimes \mathcal{P}(j) & \xrightarrow{\cong} & \mathcal{P}(j) \\
\eta \otimes \text{Id} \downarrow & \nearrow \gamma & \\
\mathcal{P}(1) \otimes \mathcal{P}(j) & &
\end{array}$$

Examples 2.2. Let us consider some examples of planar operads in a symmetric monoidal category $\mathcal{C}$:

1. The *associative operad* $\text{Ass}^\otimes$ is the operad whose algebras are associative monoids. We will define what it means to be an algebra over an operad later. However, the associative planar operad in $\mathcal{C}$ is the operad $\text{Ass}^\otimes_{\mathcal{C}}(n) = \mathbf{I}$ for all $i \in \mathbb{N}$ with $\text{Id}_{\mathbf{I}}$ being the unit and the composition map being the canonical isomorphism. Hence, for $\mathcal{C} = \mathbf{Set}$, the category of sets equipped with the cartesian symmetric monoidal structure, we obtain exactly one n -ary operation for each n : $\text{Ass}^\otimes_{\mathbf{Set}}$ is defined by $\text{Ass}^\otimes_{\mathbf{Set}}(n) = *$ for all n . Similarly, for $\mathcal{C} = \mathbf{Vect}_K$ we get the free abelian vector space over $*$, i.e. K .
2. At the beginning of the section we already saw that the concatenation of loops defines an operad: Let $\mathcal{C} = \mathbf{Top}$. Then we can define the *operad of little interval* $\mathbf{E}$ as

$$\mathbf{E}(n) = \text{Emb}(I_1, \dots, I_n, I),$$

the set of orientation-preserving embeddings of n disjoint intervals into the unit interval I . We equip $\mathbf{E}(n)$ with the subspace topology of $\mathbb{R}^n$. Composition is defined by inserting the intervals into another interval as we have seen above.

3. For $\mathcal{C}$ being closed, i.e. having an internal hom functor $\underline{\text{Hom}}(-, -)$, we can define another planar operad End_X for any object $X \in \mathcal{C}$:

$$\text{End}_X(n) := \underline{\text{Hom}}(X^{\otimes n}, X)$$

The unit morphism η is given by the image of Id_X under the adjunction:

$$\text{Hom}(X, X) \cong \text{Hom}(I, \underline{\text{Hom}}(X, X))$$

Further, the composition morphism

$$\underline{\text{Hom}}(X^{\otimes k}, X) \otimes \bigotimes_s \underline{\text{Hom}}(X^{\otimes j_s}, X) \rightarrow \underline{\text{Hom}}(X^{\otimes j}, X)$$

is given by composition:

$$\begin{array}{c}
\underline{\text{Hom}}(X^{\otimes k}, X) \otimes \bigotimes_s \underline{\text{Hom}}(X^{\otimes j_s}, X) \\
\downarrow \\
\underline{\text{Hom}}(X^{\otimes k}, X) \otimes \underline{\text{Hom}}(X^{\otimes j}, X^{\otimes k}) \\
\downarrow \\
\underline{\text{Hom}}(X^{\otimes j}, X)
\end{array}$$

We call End_X the *endomorphism operad over X* .

For another example of an operad, which we will encounter a lot throughout this thesis, we need to recall the definition of the braid group first.

Definition 2.3. For each integer $n > 2$ we define the *braid group* on n strands as the group with $n - 1$ generators $\sigma_1, \dots, \sigma_{n-1}$ satisfying the following relations:

$$\begin{array}{ll}
\sigma_i \sigma_j = \sigma_j \sigma_i & \text{for } |i - j| > 1 \\
\sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_i \sigma_{i+1} & \text{for } 1 \leq i \leq n - 2
\end{array}$$

For $n = 2$ we define B_2 as the free group with one generator and let $B_0 = B_1 = \{1\}$ be the trivial group.

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Remark 2.4. The name braid group B_n makes sense when we consider the following depiction of the group for $0 \leq i, j \leq n$ with $i+1 < j$:

$$\begin{aligned}
 \sigma_i &= \begin{array}{c} \text{Diagram of a braid with } i \text{ and } i+1 \text{ strands crossing} \end{array} \\
 \sigma_i \circ \sigma_j &= \begin{array}{c} \text{Diagram of two braid blocks: first } \sigma_i \text{ on strands } i, i+1, \text{ then } \sigma_j \text{ on strands } j, j+1 \end{array} = \sigma_j \circ \sigma_i \\
 \sigma_i \circ \sigma_{i+1} \circ \sigma_i &= \begin{array}{c} \text{Diagram of three braid blocks: } \sigma_i \text{ on } i, i+1, \text{ then } \sigma_{i+1} \text{ on } i+1, i+2, \text{ then } \sigma_i \text{ on } i, i+1 \end{array} = \begin{array}{c} \text{Diagram of three braid blocks: } \sigma_{i+1} \text{ on } i+1, i+2, \text{ then } \sigma_i \text{ on } i, i+1, \text{ then } \sigma_{i+1} \text{ on } i+1, i+2 \end{array} = \sigma_{i+1} \circ \sigma_i \circ \sigma_{i+1}
 \end{aligned}$$

Further details on how this is defined formally can be found in [CS].

We state some facts about braid groups that are known and can for example be found in [DY].

Remark 2.5. 1. There is a projection, which we denote by $\pi: B_n \rightarrow S_n$, that adds the condition $\sigma_i^2 = \text{Id}$.

2. The direct sum homomorphism

$$\bigoplus_S: S_{k_1} \times \cdots \times S_{k_n} \rightarrow S_{k_1 + \cdots + k_n}$$

can be lifted to braid groups:

$$\bigoplus_B: B_{k_1} \times \cdots \times B_{k_n} \rightarrow B_{k_1 + \cdots + k_n}$$

such that $\bigoplus_B \circ \pi = (\pi)^k \circ \bigoplus_S$. Geometrically, $b_1 \oplus \cdots \oplus b_n \in B_{k_1 + \cdots + k_n}$ describes the braid that places $b_1, \dots, b_n$ side by side from left to right.

3. For $n \geq 1$ and $k_1, \dots, k_n \in \mathbb{N}$, the block permutation map

$$(-)_S(k_1, \dots, k_n): S_n \rightarrow S_{k_1 + \cdots + k_n}$$

can be lifted to braid groups:

$$(-)_B(k_1, \dots, k_n): B_n \rightarrow B_{k_1 + \cdots + k_n}$$

such that $(-)_B(k_1, \dots, k_n) \circ \pi = \pi \circ (-)_S(k_1, \dots, k_n)$. Geometrically, for $b \in B_n$, the corresponding block braid $b(k_1, \dots, k_n)$ is obtained by replacing the i -th strand by k_i many parallel strands for $1 \leq i \leq n$.

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4. Let $\sigma, \rho \in B_n$ and $\tau_j \in B_{k_j}$ for $1 \leq j \leq n$. Then the following hold in $B_{k_1+\dots+k_n}$:

$$\begin{aligned} (\sigma\rho)(k_1, \dots, k_n) &= \sigma(k_{\bar{\rho}^{-1}(1)}, \dots, k_{\bar{\rho}^{-1}(n)}) \cdot \rho(k_1, \dots, k_n) \\ \sigma(k_1, \dots, k_n) \cdot (\tau_1 \oplus \dots \oplus \tau_n) &= (\tau_{\bar{\sigma}^{-1}(1)} \oplus \dots \oplus \tau_{\bar{\sigma}^{-1}(n)}) \cdot \sigma(k_1, \dots, k_n) \end{aligned}$$

where $\bar{\sigma} = \pi(\sigma)$ and $\bar{\rho} = \pi(\rho)$.

Example 2.6. Let $\mathcal{B}$ be the *braid operad* defined by:

$$\mathcal{B}(n) := B_n,$$

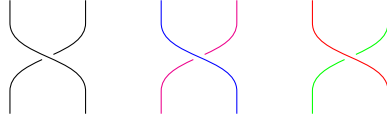
where B_n denotes the braid group on n strands. The composition morphism

$$\begin{aligned} \gamma: B_k \times B_{j_1} \times \dots \times B_{j_k} &\rightarrow B_j \\ (\tau, (\sigma_i)_{1 \leq i \leq k}) &\mapsto \tau(\sigma_1 \oplus \dots \oplus \sigma_k) \end{aligned}$$

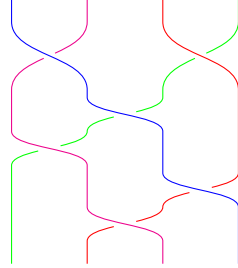
where $\tau(\sigma_1 \oplus \dots \oplus \sigma_k)$ denotes the braid that τ -braids k many block braids $\sigma_1, \dots, \sigma_k$ as if they were strands in B_k . Let us familiarize ourselves with the composition by looking at an example. For the composition map

$$B_2 \times B_2 \times B_2 \rightarrow B_4$$

we consider the following element in $B_2 \times B_2 \times B_2$:



When applying the composition map we obtain the following braiding:



The associativity and unitality conditions follow from the group conditions.

Definition 2.7. Let $\mathcal{P}, \mathcal{P}'$ be planar operads in a symmetric monoidal category $\mathcal{C}$. A *morphism of planar operads* $\beta: \mathcal{P} \rightarrow \mathcal{P}'$ consists of morphisms $\beta_n: \mathcal{P}(n) \rightarrow \mathcal{P}'(n)$ for all $n \in \mathbb{N}$ such that the following diagrams commute:

$$\begin{array}{ccc} \mathcal{P}(k) \otimes \mathcal{P}(j_1) \otimes \dots \otimes \mathcal{P}(j_k) & \xrightarrow{\beta_k \otimes \beta_{j_1} \otimes \dots \otimes \beta_{j_k}} & \mathcal{P}'(k) \otimes \mathcal{P}'(j_1) \otimes \dots \otimes \mathcal{P}'(j_k) \\ \gamma \downarrow & & \downarrow \gamma' \\ \mathcal{P}(j) & \xrightarrow{\beta_j} & \mathcal{P}'(j) \end{array}$$

$$\begin{array}{ccc} & k & \\ \eta \swarrow & & \searrow \eta' \\ \mathcal{P}(1) & \xrightarrow{\beta_1} & \mathcal{P}'(1) \end{array}$$

Therefore, for any closed symmetric monoidal category $\mathcal{C}$ the planar operads and their morphisms form a category $\mathbf{PlanarOp}_{\mathcal{C}}$.

When we later discuss operads, we will seldom focus on planar operads but will work with symmetric operads instead.

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Definition 2.8. A (*symmetric*) *operad* $\mathcal{O}$ in $\mathcal{C}$ is a planar operad as defined in definition 2.1 with an action of S_n on $\mathcal{O}(n)$ together with the following equivariance condition: Let $\sigma(j_1, \dots, j_k) \in S_j$ denote the element that permutes k blocks of length j_l for $1 \leq l \leq k$. Then, the following diagram commutes:

$$\begin{array}{ccc} \mathcal{O}(k) \otimes \mathcal{O}(j_1) \otimes \dots \otimes \mathcal{O}(j_k) & \xrightarrow{\sigma \otimes \sigma^{-1}} & \mathcal{O}(k) \otimes \mathcal{O}(j_{\sigma(1)}) \otimes \dots \otimes \mathcal{O}(j_{\sigma(k)}) \\ \gamma \downarrow & & \downarrow \gamma \\ \mathcal{O}(j) & \xrightarrow{\sigma(j_1 \dots j_k)} & \mathcal{O}(j). \end{array}$$

Furthermore, let $\tau_l \in S_{j_l}$ and $\tau_1 \oplus \dots \oplus \tau_k$ denote the sum of permutations on k blocks of length j_l for $1 \leq l \leq k$. Then the following diagram commutes:

$$\begin{array}{ccc} \mathcal{O}(k) \otimes \mathcal{O}(j_1) \otimes \dots \otimes \mathcal{O}(j_k) & \xrightarrow{\text{Id} \otimes \tau_1 \otimes \dots \otimes \tau_k} & \mathcal{O}(k) \otimes \mathcal{O}(j_1) \otimes \dots \otimes \mathcal{O}(j_k) \\ \gamma \downarrow & & \downarrow \gamma \\ \mathcal{O}(j) & \xrightarrow{\tau_1 \oplus \dots \oplus \tau_k} & \mathcal{O}(j). \end{array}$$

As already mentioned we introduce operads as a tool to generally encode algebraic structure.

Definition 2.9. Let $\mathcal{O}$ be an operad in $\mathcal{C}$. An $\mathcal{O}$ -*algebra* is an object A in $\mathcal{C}$ with maps

$$\theta: \mathcal{O}(j) \otimes A^j \rightarrow A$$

for $j \in \mathbb{N}$ such that the following associativity, unitality and equivariant conditions hold:

- Associativity:

$$\begin{array}{ccc} \mathcal{O}(k) \otimes \mathcal{O}(j_1) \otimes \dots \otimes \mathcal{O}(j_k) \otimes A^j & \xrightarrow{\gamma \otimes \text{Id}} & \mathcal{O}(j) \otimes A^j \\ \downarrow s & & \downarrow \theta \\ \mathcal{O}(k) \otimes (\bigotimes (\mathcal{O}(j_l) \otimes A^{j_l})) & \xrightarrow{\text{Id} \otimes \theta^k} & \mathcal{O}(k) \otimes A^k \\ & & \uparrow \theta \\ & & A \end{array}$$

- Unitality:

$$\begin{array}{ccc} \text{I} \otimes A & \xrightarrow{\cong} & A \\ \eta \otimes \text{Id} \downarrow & \nearrow \theta & \\ \mathcal{O}(1) \otimes A & & \end{array}$$

- Equivariance: For all $j \in \mathbb{N}$ and $\sigma \in S_j$:

$$\begin{array}{ccc} \mathcal{O}(j) \otimes A^j & \xrightarrow{\sigma \otimes \sigma^{-1}} & \mathcal{O}(j) \otimes A^j \\ & \searrow \theta & \swarrow \theta \\ & A & \end{array}$$

Definition 2.10. Let $\mathcal{O}$ be an operad in $\mathcal{C}$, A an $\mathcal{O}$ -algebra. We call an object M in $\mathcal{C}$ with maps

$$\omega: \mathcal{O}(j) \otimes A^{j-1} \otimes M \rightarrow M$$

an A -*module* if it satisfies the following associativity, unitality and equivariance conditions:

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- Associativity: For all $\sum j_l = j$ in $\mathbb{N}$ the following diagrams commute:

$$\begin{array}{ccc}
 \mathcal{O}(k) \otimes \mathcal{O}(j_1) \otimes \cdots \otimes \mathcal{O}(j_k) \otimes A^{j-1} \otimes M & \xrightarrow{\gamma \otimes \text{Id}} & \mathcal{O}(j) \otimes A^{j-1} \otimes M \\
 \downarrow s & & \downarrow \omega \\
 \mathcal{O}(k) \otimes \left(\bigotimes_{l=1}^{k-1} \mathcal{O}(j_l) \otimes A^{j_l} \right) \otimes \mathcal{O}(j_k) \otimes A^{j_k-1} \otimes M & \xrightarrow{\text{Id} \otimes \theta^{k-1} \otimes \omega} & \mathcal{O}(k) \otimes A^{k-1} \otimes M \\
 & & \uparrow \omega
 \end{array}$$

- Unitality:

$$\begin{array}{ccc}
 \text{I} \otimes M & \xrightarrow{\cong} & M \\
 \eta \otimes \text{Id} \downarrow & \nearrow \omega & \\
 \mathcal{O}(1) \otimes A^0 \otimes M & &
 \end{array}$$

- Equivariance: For all $j \in \mathbb{N}$ and $\sigma \in S_{j-1} \subset S_j$, the following diagram commutes:

$$\begin{array}{ccc}
 \mathcal{O}(j) \otimes A^{j-1} \otimes M & \xrightarrow{\sigma \otimes \sigma^{-1} \otimes \text{Id}} & \mathcal{O}(j) \otimes A^{j-1} \otimes M \\
 \searrow \omega & & \swarrow \omega \\
 & M &
 \end{array}$$

Remark 2.11. We can of course also define algebras over planar operads by dropping the equivariance condition as there is no symmetric group action. So, algebras over the operad $\text{Ass}^\otimes$ which we defined in examples 2.2 are indeed monoids since the corresponding squares commute due to canonical isomorphisms:

$$\begin{array}{ccc}
 \text{Ass}^\otimes(2) \otimes \text{Ass}^\otimes(2) \otimes \text{Ass}^\otimes(1) \otimes A^3 & \xrightarrow{\cong} & \text{Ass}^\otimes(3) \otimes A^3 \\
 \downarrow \cong & & \downarrow \theta \\
 \text{Ass}^\otimes(2) \otimes ((\text{Ass}^\otimes(2) \otimes A^2) \otimes (\text{Ass}^\otimes(1) \otimes A)) & \xrightarrow{\text{Id} \otimes \theta^2} & \text{Ass}^\otimes(2) \otimes A^2 \\
 & & \uparrow \theta \\
 \text{Ass}^\otimes(2) \otimes \text{Ass}^\otimes(1) \otimes \text{Ass}^\otimes(2) \otimes A^3 & \xrightarrow{\cong} & \text{Ass}^\otimes(3) \otimes A^3 \\
 \downarrow \cong & & \downarrow \theta \\
 \text{Ass}^\otimes(2) \otimes ((\text{Ass}^\otimes(1) \otimes A) \otimes (\text{Ass}^\otimes(2) \otimes A^2)) & \xrightarrow{\text{Id} \otimes \theta^2} & \text{Ass}^\otimes(2) \otimes A^2 \\
 & & \uparrow \theta
 \end{array}$$

Examples 2.12. 1. A natural question is whether the associative planar operad and the associative operad in a category $\mathcal{C}$ coincide. For $\mathcal{C} = \mathbf{Set}$, we can equip the associative operad

$$\text{Ass}^\otimes_{\mathbf{Set}}^{\text{planar}}(n) = *$$

with the trivial action of S_n . Yet this appears to be too restrictive: The equivariance condition implies that algebras over this operad are not just monoids but commutative monoids. To distinguish operations after acting by S_n , we can think of a more suited candidate: $\text{Ass}^\otimes(n) = S_n$ for all $n \in \mathbb{N}$ where S_n acts on $\text{Ass}^\otimes(n)$ by translation. The

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composition map is defined by considering a sum of permutations and then permuting k blocks of length j_i for $1 \leq i \leq k$:

$$\begin{aligned} \gamma: \text{Ass}^\otimes(k) \times \text{Ass}^\otimes(j_1) \times \cdots \times \text{Ass}^\otimes(j_k) &\rightarrow \text{Ass}^\otimes(k) \\ (\sigma, \tau_1, \dots, \tau_k) &\mapsto \sigma(\tau_1 \oplus \cdots \oplus \tau_k) \end{aligned}$$

We might also refer to this operad as the symmetry operad $\mathcal{S}$ later on.

2. The operad $\text{Comm}^\otimes_{\mathbf{Set}}(n) = *$ with the trivial action of S_n is an interesting operad nevertheless as it is the *commutative operad* in the category $\mathbf{Set}$. The commutative operad in a category $\mathcal{C}$ is the operad such that its algebras are commutative monoids in $\mathcal{C}$.
3. The *little n -disk operad* D_n is an important example of an operad in the category of topological spaces **Top**. For all $n \in \mathbb{N}$ we will define $D_n(k)$ as a topological space representing the space of k disjoint little disks inside the unit disk in the plane. Let $n \in \mathbb{N}$ and

$$D^n = \{(r_1, \dots, r_n) \in \mathbb{R}^n : r_1^2 + \cdots + r_n^2 \leq 1\} \subset \mathbb{R}^n$$

is the closed unit n -disk. A *little n -disk* is defined as an embedding $f: D^n \rightarrow D^n$ of the following form:

$$f(r_1, \dots, r_n) := (c_1, \dots, c_n) + \lambda(r_1, \dots, r_n)$$

for $(c_1, \dots, c_n) \in D^n$ and $\lambda \in \mathbb{R}$ such that

$$0 < \lambda \leq 1 - \sum_{i=1}^n c_i^2.$$

We denote the interior of a little disk by $\text{int}(f)$ and define for $k > 0$:

$$D_n(k) := \left\{ f_1 \sqcup \cdots \sqcup f_n \in C\left(\bigsqcup_{i=1}^k D^n, D^n\right) : \text{int}(f_i) \cap \text{int}(f_j) = \emptyset \text{ for all } i \neq j \right\}$$

We equip $C\left(\bigsqcup_{i=1}^k D^n, D^n\right)$ with the compact-open topology and $D_n(k)$ with the subspace topology. $D_n(0)$ is the one point space. There is an operad structure to be defined on

$$D_n = \{D_n(k)\}_{k \geq 0}.$$

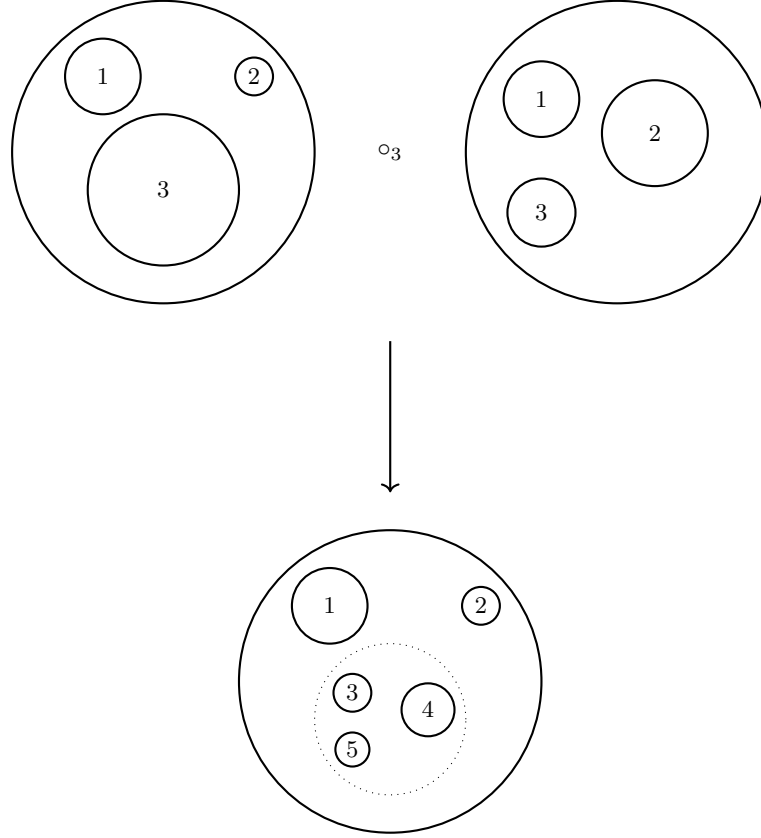
The construction of the composition map

$$\gamma: D_n(k) \times D_n(l_1) \times \cdots \times D_n(l_k) \rightarrow D_n(l_1 + \cdots + l_k)$$

is geometrically interpreted as:

- Start with a configuration in $D_n(k)$, which gives k disks inside the unit disk.
- Each of the k disks is treated as a new "unit disk" in which a configuration from $D_n(l_1), D_n(l_2), \dots, D_n(l_k)$ can be inserted.
- The configurations from $D_n(l_1), D_n(l_2), \dots, D_n(l_k)$ are scaled down to fit inside the respective disks from the configuration in $D_n(k)$, and they are inserted.
- After placing all configurations inside the k disks, the boundaries of the k disks are erased, yielding a new configuration in $D_n(n_1 + \cdots + n_k)$.

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Formally, this means that for $(f, (g_i)_{1 \leq i \leq k}) \in D_n(l_1) \times D_n(l_2) \times \cdots \times D_n(l_k)$ we define $\gamma((f, (g_i)_{1 \leq i \leq k}))$ via the following commuting diagram:

$$\begin{array}{ccc}
 \bigsqcup_{j=1}^l D^n & \xrightarrow{\gamma(f, (g_i)_{1 \leq i \leq k})} & D^n \\
 \downarrow \cong & & \uparrow f \\
 \bigsqcup_{i=1}^k \left(\bigsqcup_{p=1}^{l_k} D^n \right) & \xrightarrow{\bigsqcup g_i} & \bigsqcup_{i=1}^k D^n
 \end{array}$$

The unit is the identity configuration, where the single disk remains unchanged, i.e. the identity map $\text{Id}_{D^n} \in C(D^n, D^n)$. This satisfies the unitality condition for operads by definition, where inserting a single disk into another disk does not alter any configuration: $\gamma((f, (\text{Id})_{1 \leq i \leq k})) = f$ and $\gamma((\text{Id}, g)) = g$.

The composition map satisfies the associativity condition by definition.

The disks are labeled, and there is an action of the symmetric group S_k on $D_n(k)$, which permutes the labels of the disks: For $f_1 \sqcup \cdots \sqcup f_k \in D_n(k)$ and $\sigma \in S_k$:

$$\sigma(f_1 \sqcup \cdots \sqcup f_k) := f_{\sigma(1)} \sqcup \cdots \sqcup f_{\sigma(k)}$$

The composition map γ respects the action of S_k : For any $\sigma \in S_k$, $\tau_i \in S_{l_i}$ and

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$(f, (g_i)_{1 \leq i \leq k}) \in D_n(k) \times D_n(l_1) \times \cdots \times D_n(l_k)$ we have by definition

$$\begin{aligned} \sigma \circ \gamma((f, (g_i)_{1 \leq i \leq k})) &= \sigma \left(\bigsqcup_{i=1}^k f_i \circ g_i \right) \\ &= \bigsqcup_{i=1}^k f_{\sigma(i)} \circ g_{\sigma(i)} \\ &= \gamma((f_{\sigma(1)} \sqcup \cdots \sqcup f_{\sigma(k)}, g_{\sigma(1)}, \dots, g_{\sigma(k)}) \\ &= \gamma \circ (\sigma \times \sigma^{-1})((f, (g_i)_{1 \leq i \leq k})) \end{aligned}$$

and

$$\begin{aligned} (\tau_1 \oplus \cdots \oplus \tau_k) \circ \gamma((f, (g_i)_{1 \leq i \leq k})) &= (\tau_1 \oplus \cdots \oplus \tau_k) \left(\bigsqcup_{i=1}^k f_i \circ g_i \right) \\ &= \bigsqcup_{i=1}^k f_i \circ \tau_i(g_i) \\ &= \gamma((f, \tau_1(g_1), \dots, \tau_k(g_k))) \\ &= \gamma \circ (\tau_1 \times \cdots \times \tau_k)((f, (g_i)_{1 \leq i \leq k})). \end{aligned}$$

The little n -disk operad D_n is a model of the E_n operad and specifically E_2 will be of great importance to us throughout the entire thesis. There are other models for the E_n operads. In fact there are recognition principles for E_n operads, which will be the motivation for section 3. However, we will not discuss these other models here. So, when we talk about E_n we are referring to the little n -disk operad D_n .

2.2 Colored Operads

Following [JL], the goal of this section is to introduce colored operads in such a way that we can define colored ∞ -operads later on. A symmetric monoidal category is an example of a colored operad. Thus, we will focus on this special case first. We will show that a symmetric monoidal category can be defined in a purely functorial way. After that we will then follow this approach and define the notion of a colored operad in a similar way.

Let us start by introducing some terminology from [Kerodon].

Definition 2.13. Let $P: \mathcal{C} \rightarrow \mathcal{D}$ be a functor of arbitrary categories and $f: x \rightarrow y$ a morphism in $\mathcal{C}$. We call f *cartesian* if for any morphism $g: z \rightarrow y$ in $\mathcal{C}$ and $\omega: P(z) \rightarrow P(x)$ in $\mathcal{D}$ with $P(f) \circ \omega = P(g)$, there exists a unique morphism $\omega': z \rightarrow x$ in $\mathcal{C}$ such that $g = f \circ \omega'$ and $P(\omega') = \omega$. In diagrams we have:

$$\begin{array}{ccc} \begin{array}{ccc} z & \xrightarrow{g} & y \\ \exists! \omega' \downarrow & \nearrow f & \\ x & & \end{array} & \xrightarrow{P} & \begin{array}{ccc} P(z) & \xrightarrow{P(g)} & P(y) \\ \omega \downarrow & \nearrow P(f) & \\ P(x) & & \end{array} \end{array}$$

And vice versa, we call f *cocartesian* if we can flip arrows of the above case, i.e. for any morphism $g: y \rightarrow z$ in $\mathcal{C}$ and $\omega: P(x) \rightarrow P(z)$ in $\mathcal{D}$ with $\omega \circ P(f) = P(g)$, there is a unique morphism $\omega': x \rightarrow z$ such that $P(\omega') = \omega$ and $\omega' \circ f = g$. In a diagram this means the following:

$$\begin{array}{ccc} \begin{array}{ccc} y & \xrightarrow{g} & z \\ & \searrow f & \uparrow \exists! \omega' \\ & & x \end{array} & \xrightarrow{P} & \begin{array}{ccc} P(y) & \xrightarrow{P(g)} & P(z) \\ & \searrow P(f) & \uparrow \omega \\ & & P(x) \end{array} \end{array}$$

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Note that equivalently we can define a cocartesian morphism by requiring a bijection:

$$\mathrm{Hom}_{\mathcal{C}}(y, z) \times_{\mathrm{Hom}_{\mathcal{D}}(P(y), P(z))} \mathrm{Hom}_{\mathcal{D}}(P(x), P(z)) \cong \mathrm{Hom}_{\mathcal{C}}(x, z)$$

where $\mathrm{Hom}_{\mathcal{C}}(y, z) \times_{\mathrm{Hom}_{\mathcal{D}}(P(y), P(z))} \mathrm{Hom}_{\mathcal{D}}(P(x), P(z))$ is the pullback of applying the functor P to $\mathrm{Hom}_{\mathcal{C}}(y, z)$ and precomposing with $P(f): P(y) \rightarrow P(x)$. So, for the bijection we send a pair (g, ω) to the unique morphism ω' and vice versa we send a morphism $h: x \rightarrow z$ to the pair $(h \circ f, P(h))$.

Definition 2.14. Let $P: \mathcal{C} \rightarrow \mathcal{D}$ be a functor. P is called a *Grothendieck fibration* if for every object y in $\mathcal{C}$ and morphisms $f': x' \rightarrow P(y)$ there exists a cartesian morphism $f: x \rightarrow y$ such that $f' = P(f)$. We call f a cartesian lift of f' and f' a cleavage. Dually we define the notion of a *Grothendieck op-fibration*, i.e. P is a Grothendieck op-fibration if for every object x in $\mathcal{C}$ and morphism $f': P(x) \rightarrow y'$ there exists a cocartesian morphism $f: x \rightarrow y$ such that $P(f) = f'$.

If not stated otherwise, when we talk about fibrations and op-fibrations in this thesis, we will always mean Grothendieck fibrations and Grothendieck op-fibrations, respectively. Also note, that in [Kerodon], these terms are referred to as cartesian and cocartesian fibration. However, we chose to use the terminology used in [JL].

In the next part we will construct a functor having certain properties from a symmetric monoidal category. We will then see that the functor encodes all the necessary data of the symmetric monoidal category.

Definition 2.15. Let $\mathcal{C}$ be a symmetric monoidal category. We define a new category $\mathcal{C}^{\otimes}$ in the following way:

- Objects of $\mathcal{C}^{\otimes}$ are finite sequences $[C_1, \dots, C_n]$ where C_i is an object in $\mathcal{C}$ for all i . It is possible that the sequence is empty.
- A morphism $f: [C_1, \dots, C_n] \rightarrow [C'_1, \dots, C'_m]$ in $\mathcal{C}^{\otimes}$ consists of a subset $S_f \subset \{1, \dots, n\}$, a map of finite sets $\alpha_f: S_f \rightarrow \{1, \dots, m\}$ and a family of morphisms

$$\left\{ f_j: \bigotimes_{\alpha_f(i)=j} C_i \rightarrow C'_j \right\}_{1 \leq j \leq m}$$

in $\mathcal{C}$.

- Composition is defined as follows: Let

$$f: [C_1, \dots, C_n] \rightarrow [C'_1, \dots, C'_m] \quad \text{and} \quad g: [C'_1, \dots, C'_m] \rightarrow [C''_1, \dots, C''_k]$$

be morphisms in $\mathcal{C}^{\otimes}$, then

$$g \circ f: [C_1, \dots, C_n] \rightarrow [C''_1, \dots, C''_k]$$

is defined via $S_{g \circ f} = \alpha_f^{-1}(S_g)$, $\alpha_{g \circ f} = \alpha_g \circ \alpha_f|_{S_{g \circ f}}$ and for all $1 \leq l \leq k$ we have:

$$\bigotimes_{\alpha_{g \circ f}(i)=l} C_i \cong \bigotimes_{\alpha_g(j)=k} \bigotimes_{\alpha_f(i)=j} C_i \rightarrow \bigotimes_{\alpha_g(j)=k} C'_j \rightarrow C''_l.$$

Let I be a finite set. We write $I_* = I \amalg *$.

Definition 2.16. We define Fin_* , the *category of finite pointed sets*, as follows:

- For each $n \in \mathbb{N}$, we write $\langle n \rangle^\circ := \{1, \dots, n\}$. We define the objects of Fin_* as $\langle n \rangle := \langle n \rangle_*^\circ$.
- A morphism $\alpha: \langle n \rangle \rightarrow \langle m \rangle$ is a map of finite sets where $\alpha(*) = *$.

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It is clear that this defines a category. Further note that this category is equivalent to the category of finite sets with a distinguished point since for any finite set I we have an isomorphism $I_* \cong \langle n \rangle$, where n is the cardinality of I . For all pairs of $1 \leq i \leq n$ we write $\rho^i: \langle n \rangle \rightarrow \langle 1 \rangle$ for the morphism in $\mathcal{F}\text{in}_*$ which is defined by

$$\rho^i(j) = \begin{cases} 1 & i = j \\ * & \text{else.} \end{cases}$$

For any symmetric monoidal category $\mathcal{C}$ there exists a forgetful functor

$$P: \mathcal{C}^\otimes \rightarrow \mathcal{F}\text{in}_*$$

which sends a sequence $[C_1, \dots, C_n]$ to $\langle n \rangle$ and a morphism $f: [C_1, \dots, C_n] \rightarrow [C'_1, \dots, C'_m]$ to $\alpha: \langle n \rangle \rightarrow \langle m \rangle$ with

$$\alpha(x) = \begin{cases} \alpha_f(x) & x \in S_f \\ * & \text{else.} \end{cases}$$

Remark 2.17. Let $\mathcal{C}_0$ denote the trivial category of one object and one morphism. We claim that the forgetful functor $P: \mathcal{C}_0^\otimes \rightarrow \mathcal{F}\text{in}_*$ is an isomorphism of categories. This holds, since for all morphisms in $\mathcal{C}_0^\otimes$ the family of morphisms f_j in $\mathcal{C}_0$ is trivial, in other words there is nothing to forget. Hence, $\mathcal{F}\text{in}_*$ is obtained as the simplest example of the construction of a $\mathcal{C}^\otimes$ category. Furthermore, we note that the forgetful functor $P: \mathcal{C}^\otimes \rightarrow \mathcal{F}\text{in}_*$ can be obtained from a unique symmetric monoidal functor $F: \mathcal{C} \rightarrow \mathcal{C}_0$ using the functoriality of F .

Remark 2.18. For every object $\langle n \rangle$ we denote the fiber category by $\mathcal{C}_{\langle n \rangle}$, which is defined to be the pullback of the following diagram

$$\begin{array}{ccc} \mathcal{C}_{\langle n \rangle}^\otimes & \dashrightarrow & \mathcal{C}^\otimes \\ \downarrow & & \downarrow P \\ \{\langle n \rangle\} & \hookrightarrow & \mathcal{F}\text{in}_* \end{array}$$

where $\{\langle n \rangle\}$ is interpreted as the trivial one object category. In other words, $\mathcal{C}_{\langle n \rangle}$ consists of all collections of objects of length n and all morphisms f such that $\alpha_f = \text{Id}_{\langle n \rangle}$.

We are now ready to state two very important properties of the functor $P: \mathcal{C}^\otimes \rightarrow \mathcal{F}\text{in}_*$, which as it turns out is all we need to know.

Proposition 2.19. *Let $P: \mathcal{C}^\otimes \rightarrow \mathcal{F}\text{in}_*$ be the forgetful functor as describes above. Then, the following conditions hold:*

(M1) **Op-fibration condition:** *P is a Grothendieck op-fibration.*

(M2) **Segal condition:** *We have that $\mathcal{C} \cong \mathcal{C}_{\langle 1 \rangle}^\otimes$ and more precisely we obtain that $\mathcal{C}_{\langle n \rangle}^\otimes \cong \mathcal{C}^n$ where $\mathcal{C}^n$ denotes the n -fold product of copies of $\mathcal{C}$.*

Proof. (M1) Let $[C_1, \dots, C_n]$ be an object in $\mathcal{C}^\otimes$ and $f': \langle n \rangle \rightarrow \langle m \rangle$ a morphism in $\mathcal{F}\text{in}_*$. We need to construct a cocartesian morphism $f: [C_1, \dots, C_n] \rightarrow [C'_1, \dots, C'_m]$ such that $P(f) = f'$. We define $\alpha_f: S_f \rightarrow \langle m \rangle^\circ$ as $\alpha_f := f'|_{S_f}$ where $S_f := f'^{-1}(\langle m \rangle^\circ)$. Further we define a selection of objects in $\mathcal{C}$ for $1 \leq j \leq m$:

$$C'_j := \begin{cases} \bigotimes_{\alpha_f(i)=j} C_i & j \in \alpha_f(S_f) \\ \text{I} & \text{else.} \end{cases}$$

For $f_j := \text{Id}_{C'_j}$ we have then constructed a well defined morphism

$$f: [C_1, \dots, C_n] \rightarrow [C'_1, \dots, C'_m]$$

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such that $P(f) = f'$. We are left to show that f is indeed cocartesian. Therefore, let $g: [C_1, \dots, C_n] \rightarrow [C'_1, \dots, C'_l]$ be a morphism in $\mathcal{C}^\otimes$ and $\omega': \langle m \rangle \rightarrow \langle l \rangle$ a morphism in $\mathcal{F}\text{in}_*$ such that $P(g) = \omega' \circ P(f)$. We need to find a unique morphism

$$\omega: [C'_1, \dots, C'_m] \rightarrow [C''_1, \dots, C''_l]$$

such that $g = \omega \circ f$ and $P(\omega) = \omega'$. We are forced to define $\alpha_\omega := \omega'|_{S_\omega}$ where

$$S_\omega = \omega'^{-1}(\langle l \rangle^\circ).$$

By construction we have $\alpha_g = \alpha_\omega \circ \alpha_f$ and by definition of C'_j we are forced to define for all $1 \leq j \leq l$:

$$\omega_j: \bigotimes_{\alpha_\omega(k)=j} C'_k \cong \bigotimes_{\alpha_\omega(k)=j} \bigotimes_{\alpha_f(i)=k} C_i \cong \bigotimes_{\alpha_g(k)=j} C_k \xrightarrow{g_j} C''_j$$

Uniqueness of ω follows from $\mathcal{C}$ being a symmetric monoidal category. Hence, f is cocartesian.

(M2) Let $F: \mathcal{C}_{\langle 1 \rangle}^\otimes \rightarrow \mathcal{C}$ be a functor that sends $[C]$ to C . Let $f: [C] \rightarrow [C']$ be a morphism in $\mathcal{C}_{\langle 1 \rangle}^\otimes$, then $\alpha_f = \text{Id}_{\{1\}}$ and there is just one morphism $f_1: C \rightarrow C'$ corresponding to f . So, we define $F(f) = f_1$. It follows by definition that F is fully faithful and essentially surjective, i.e. an equivalence of categories.

First, we will show that for any morphism $\langle n \rangle \rightarrow \langle m \rangle$ the op-fibration condition (M1) induces a functor of fiber categories $A: \mathcal{C}_{\langle n \rangle}^\otimes \rightarrow \mathcal{C}_{\langle m \rangle}^\otimes$: Let $\alpha: \langle n \rangle \rightarrow \langle m \rangle$ be a morphism in $\mathcal{F}\text{in}_*$ and $[C_1, \dots, C_n]$ an object in $\mathcal{C}_{\langle n \rangle}^\otimes$. By the op-fibration condition (M1), we can find a cocartesian morphism

$$f_\alpha: [C_1, \dots, C_n] \rightarrow [C'_1, \dots, C'_m].$$

Therefore, we can define

$$A([C_1, \dots, C_n]) := [C'_1, \dots, C'_m].$$

To see that the choice of $[C'_1, \dots, C'_m]$ is unique up to isomorphism, let

$$f'_\alpha: [C_1, \dots, C_n] \rightarrow [\tilde{C}'_1, \dots, \tilde{C}'_m]$$

be a different choice of cocartesian morphism for $[C_1, \dots, C_n]$ and α . Then there are morphisms

$$g: [C'_1, \dots, C'_m] \rightarrow [\tilde{C}'_1, \dots, \tilde{C}'_m] \quad \text{and} \quad g': [\tilde{C}'_1, \dots, \tilde{C}'_m] \rightarrow [C'_1, \dots, C'_m]$$

corresponding to the cocartesian property of f_α and f'_α respectively. Because of uniqueness we obtain $g \circ g' = \text{Id}_{[\tilde{C}'_1, \dots, \tilde{C}'_m]}$ and $g' \circ g = \text{Id}_{[C'_1, \dots, C'_m]}$. Therefore, the choice of a cocartesian morphism is unique up to isomorphism.

Let $f: [C_1, \dots, C_n] \rightarrow [\bar{C}_1, \dots, \bar{C}_n]$ be a morphism in $\mathcal{C}_{\langle n \rangle}^\otimes$. We obtain a unique morphism $f': [\bar{C}_1, \dots, \bar{C}_n] \rightarrow [\bar{C}'_1, \dots, \bar{C}'_m]$ from the fact that f_α is cocartesian.

$$\begin{array}{ccc} [C_1, \dots, C_n] & \xrightarrow{f_\alpha} & [C'_1, \dots, C'_m] \\ \downarrow f & & \downarrow f' \\ [\bar{C}_1, \dots, \bar{C}_n] & \xrightarrow{\bar{f}_\alpha} & [\bar{C}'_1, \dots, \bar{C}'_m] \end{array} \quad \xrightarrow{P} \quad \begin{array}{ccc} \langle n \rangle & \xrightarrow{\alpha} & \langle m \rangle \\ \downarrow \text{Id} & & \downarrow \text{Id} \\ \langle n \rangle & \xrightarrow{\alpha} & \langle m \rangle \end{array}$$

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Hence, we can define $A(f) := f'$. Due to uniqueness of f' , A preserves composition and maps identities to identities.

Let $P^i: \mathcal{C}_{\langle n \rangle}^{\otimes} \rightarrow \mathcal{C}_{\langle 1 \rangle}^{\otimes} \cong \mathcal{C}$ be the functor corresponding to $\rho^i: \langle n \rangle \rightarrow \langle 1 \rangle$ which we defined previously for all $1 \leq i \leq n$. Then from the universal property of the product, it follows that

$$\prod_{1 \leq i \leq n} P^i: \mathcal{C}_{\langle n \rangle}^{\otimes} \rightarrow \mathcal{C}^n$$

is an equivalence of categories. □

Note that in the following we might refer to the constructed functor $\mathcal{C}_{\langle n \rangle}^{\otimes} \rightarrow \mathcal{C}_{\langle m \rangle}^{\otimes}$ originated from $\alpha: \langle n \rangle \rightarrow \langle m \rangle$ also by α .

We are now well prepared to see that the symmetric monoidal structure on $\mathcal{C}$ is encoded in $\mathcal{C}^{\otimes}$ and its forgetful functor $P: \mathcal{C}^{\otimes} \rightarrow \mathcal{Fin}_*$ satisfies the above conditions. In other words, for any category $\mathcal{D}$ together with a functor $P: \mathcal{D} \rightarrow \mathcal{Fin}_*$ satisfying the op-fibration condition (M1) and the Segal condition (M2), we can construct a symmetric monoidal category $\mathcal{C}$ such that $\mathcal{D} \cong \mathcal{C}^{\otimes}$ and P is the usual forgetful functor. The right candidate for $\mathcal{C}$ is clearly $\mathcal{D}_{\langle 1 \rangle}$.

Proposition 2.20. *The construction of $\pi: \mathcal{C}^{\otimes} \rightarrow \mathcal{Fin}_*$ from a symmetric monoidal category $\mathcal{C}$ given above is a bijection:*

$$\{\text{symmetric monoidal categories}\} \cong \{\text{op-fibrations } \mathcal{D} \rightarrow \mathcal{Fin}_* \text{ satisfying (M2)}\}$$

Proof. Let $\mathcal{D}$ be a category and $P: \mathcal{D} \rightarrow \mathcal{Fin}_*$ a functor satisfying the op-fibration condition (M1) and the Segal condition (M2). We will equip $\mathcal{C} := \mathcal{D}_{\langle 1 \rangle}$ with a symmetric monoidal structure and see that there is an equivalence of categories $\mathcal{C}^{\otimes} \cong \mathcal{D}$.

1. For the map $\alpha: \langle 2 \rangle \rightarrow \langle 1 \rangle$ with $\alpha^{-1}(1) = \{1, 2\}$ we obtain the following functor using (M2):

$$\bigotimes: \mathcal{C} \times \mathcal{C} \cong \mathcal{D}_{\langle 2 \rangle} \xrightarrow{\alpha} \mathcal{C}$$

2. There is a unique morphism $\eta: \langle 0 \rangle \rightarrow \langle 1 \rangle$ in $\mathcal{Fin}_*$ defining a functor $\mathcal{D}_{\langle 0 \rangle} \rightarrow \mathcal{C}$ which gives us a unique object I in $\mathcal{C}$. This object I will serve as the tensor unit which follows from the following commuting diagram:

$$\begin{array}{ccccc} \mathcal{D}_{\langle 0 \rangle} \times \mathcal{D}_{\langle 1 \rangle} & \xrightarrow{\eta \times \text{Id}} & \mathcal{D}_{\langle 1 \rangle} \times \mathcal{D}_{\langle 1 \rangle} & \xrightarrow{\cong} & \mathcal{D}_{\langle 2 \rangle} \\ \uparrow \cong & & & & \searrow \otimes \\ \mathcal{C} & & & & \mathcal{D}_{\langle 1 \rangle} \\ \downarrow \cong & & & & \nearrow \otimes \\ \mathcal{D}_{\langle 1 \rangle} \times \mathcal{D}_{\langle 0 \rangle} & \xrightarrow{\text{Id} \times \eta} & \mathcal{D}_{\langle 1 \rangle} \times \mathcal{D}_{\langle 1 \rangle} & \xrightarrow{\cong} & \mathcal{D}_{\langle 2 \rangle} \end{array}$$

3. Furthermore, we need a symmetric braiding. This we obtain from the morphism

$$\begin{aligned} \beta: \langle 2 \rangle &\rightarrow \langle 2 \rangle \\ 1 &\mapsto 2 \\ 2 &\mapsto 1. \end{aligned}$$

We note that $\alpha \circ \beta = \alpha$ and $\rho^i \beta = \rho^{\beta(i)}$. So, we obtain the following commuting diagram:

$$\begin{array}{ccccc} \mathcal{D}_{\langle 1 \rangle} \times \mathcal{D}_{\langle 1 \rangle} & \xleftarrow{\rho^1 \times \rho^2} & \mathcal{D}_{\langle 2 \rangle} & \xrightarrow{\alpha} & \mathcal{D}_{\langle 1 \rangle} \\ & \swarrow \rho^2 \times \rho^1 & \downarrow \beta & \nearrow \alpha & \\ & & \mathcal{D}_{\langle 2 \rangle} & & \end{array}$$

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Hence, $A \otimes B \cong B \otimes A$ holds.

4. The associator corresponds to the following maps in $\mathcal{Fin}_*$:

$$\tau_i^n: \langle n \rangle \rightarrow \langle n-1 \rangle$$

$$j \mapsto \begin{cases} j & 1 \leq j \leq i \\ j-1 & i < j \leq n \\ * & j = * \end{cases}$$

So, the following commuting diagram

$$\begin{array}{ccc} \langle 3 \rangle & \xrightarrow{\tau_1^3} & \langle 2 \rangle \\ \tau_2^3 \downarrow & & \downarrow \tau_1^2 \\ \langle 2 \rangle & \xrightarrow{\tau_1^2} & \langle 1 \rangle \end{array}$$

translates to $(A \otimes B) \otimes C \cong A \otimes (B \otimes C)$. We need to see the pentagon identities:

$$\begin{array}{ccccc} & & (A \otimes B) \otimes (C \otimes D) & & \\ & \swarrow & & \searrow & \\ A \otimes (B \otimes (C \otimes D)) & & & & ((A \otimes B) \otimes C) \otimes D \\ \uparrow & & & & \downarrow \\ A \otimes ((B \otimes C) \otimes D) & \longleftarrow & & \longrightarrow & (A \otimes (B \otimes C)) \otimes D \end{array}$$

The commutativity of the following diagram in $\mathcal{Fin}_*$ translates to the pentagon identities:

$$\begin{array}{ccccccc} & & \langle 3 \rangle & \xrightarrow{\tau_2^3} & \langle 2 \rangle & & \\ & \nearrow \tau_1^4 & & \searrow \tau_1^3 & & \searrow \tau_1^2 & \\ \langle 4 \rangle & \xrightarrow{\tau_2^4} & \langle 3 \rangle & \xrightarrow{\tau_1^3} & \langle 2 \rangle & \xrightarrow{\tau_1^2} & \langle 1 \rangle \\ & \searrow \tau_3^4 & & \searrow \tau_2^3 & & \nearrow \tau_1^2 & \\ & & \langle 3 \rangle & \xrightarrow{\tau_2^3} & \langle 2 \rangle & & \end{array}$$

$$\begin{array}{ll} \tau_1^2 \circ \tau_2^3 \circ \tau_1^4 & \rightsquigarrow (A \otimes B) \otimes (C \otimes D) \\ \tau_1^2 \circ \tau_1^3 \circ \tau_1^4 & \rightsquigarrow ((A \otimes B) \otimes C) \otimes D \\ \tau_1^2 \circ \tau_1^3 \circ \tau_2^4 & \rightsquigarrow (A \otimes (B \otimes C)) \otimes D \\ \tau_1^2 \circ \tau_2^3 \circ \tau_2^4 & \rightsquigarrow A \otimes ((B \otimes C) \otimes D) \\ \tau_1^2 \circ \tau_2^3 \circ \tau_3^4 & \rightsquigarrow A \otimes (B \otimes (C \otimes D)) \end{array}$$

This construction is in bijection to the previous given construction. $\square$

Remark 2.21. With some adjustments this construction can also be done for a monoidal category $\mathcal{C}$. The category $\mathcal{C}^\otimes$ is defined analogously to the symmetric case. But we choose ordered indexing sets $[n]^\circ := \{0 < \dots < n\}$ and for a morphism $f: [C_1, \dots, C_n] \rightarrow [C'_1, \dots, C'_m]$ we require that α_f preserves the order. Further, we can define the *pointed simplex category* Δ_* as follows:

- For each $n \in \mathbb{N}$ we define an object in Δ_* as $[n] := [n]^\circ \sqcup \{*\}$.

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- The single pointed set $\{*\}$ is an object in Δ_* . There are morphisms $\{*\} \hookrightarrow [n]$ for all n .
- A morphism $\alpha: [n] \rightarrow [m]$ is a order preserving map on $\alpha^{-1}([m])$ and $\alpha(*) = *$.

Note that for all pairs $0 \leq i \leq n$, we have $\rho^i: [n] \rightarrow [0]$ in Δ_* with

$$\rho^i(j) = \begin{cases} 0 & i = j \\ * & \text{else.} \end{cases}$$

Similarly to the symmetric case, we obtain a forgetful functor:

$$P: \mathcal{C}^\otimes \rightarrow \Delta_*$$

This functor is an op-fibration. The proof in Proposition 2.19 applies. Note that for the proof the fact that morphisms in Δ_* preserve the order is essential, otherwise we could not define

$$\omega_j: \bigotimes_{\alpha_\omega(k)=j} C'_k \cong \bigotimes_{\alpha_\omega(k)=j} \bigotimes_{\alpha_f(i)=k} C_i \cong \bigotimes_{\alpha_g(k)=j} C_k \xrightarrow{g_j} C''_j$$

in a monoidal category $\mathcal{C}$. Further, the Segal condition (M2) holds as it is a result of the op-fibration condition (M1) and the fact that ρ^i is a morphism in Δ_* . It then also follows that this similar construction is again a bijection

$$\{\text{monoidal categories}\} \cong \{\text{op-fibration } \mathcal{D} \rightarrow \Delta_* \text{ satisfying (M2)}\}.$$

The proof for this is analogous to Proposition 2.20 as all morphisms in Fin_* that induce a monoidal structure on $\mathcal{C}$ are order preserving and can be considered in Δ_* .

We will now introduce the notion of a colored operad and will move on to see that, similarly to describing a symmetric monoidal structure via a functor satisfying the op-fibration condition (M1) and Segal condition (M2), there is also a functor satisfying similar properties that encode the datum of a colored operad.

Definition 2.22. A *colored operad* $\mathcal{O}$ consists of the following data:

- A collection $\{X, Y, Z; \dots\}$ which we call objects or colors of the operad.
- For every finite collection of objects $\{X_i\}_{i \in I}$ and object Y in $\mathcal{O}$, there is a set $\text{Mul}_{\mathcal{O}}(\{X_i\}_{i \in I}, Y)$ called the set of morphisms from $\{X_i\}_{i \in I}$ to Y .
- For any finite map $I \rightarrow J$ with fibers denoted by I_j for all $j \in J$, collections of objects $\{X_i\}_{i \in I}$ and $\{Y_j\}_{j \in J}$ and object Z , there is a composition map

$$\prod_{j \in J} \text{Mul}_{\mathcal{O}}(\{X_i\}_{i \in I_j}, Y_j) \times \text{Mul}_{\mathcal{O}}(\{Y_j\}_{j \in J}, Z) \rightarrow \text{Mul}_{\mathcal{O}}(\{X_i\}_{i \in I}, Z)$$

- There is a collection of morphisms $\{\text{Id}_X \in \text{Mul}_{\mathcal{O}}(\{X\}, X)\}_{X \in \mathcal{O}}$ being a left and right unit in the composition, i.e. the following diagrams commute:

$$\begin{array}{ccc} \text{Mul}_{\mathcal{O}}(X_I, Y) & \xrightarrow{\cong} & \text{Mul}_{\mathcal{O}}(X_I, Y) \times \{\text{Id}_Y\} \\ \downarrow \text{Id} & & \downarrow \\ \text{Mul}_{\mathcal{O}}(X_I, Y) & \longleftarrow & \text{Mul}_{\mathcal{O}}(X_I, Y) \times \text{Mul}_{\mathcal{O}}((Y), Y) \end{array}$$

$$\begin{array}{ccc} \text{Mul}_{\mathcal{O}}(X_I, Y) & \xrightarrow{\cong} & (\prod_{i \in I} \{\text{Id}_{X_i}\}) \times \text{Mul}_{\mathcal{O}}(X_I, Y) \\ \downarrow \text{Id} & & \downarrow \\ \text{Mul}_{\mathcal{O}}(X_I, Y) & \longleftarrow & (\prod_{i \in I} \text{Mul}_{\mathcal{O}}((X_i), X_i)) \times \text{Mul}_{\mathcal{O}}(X_I, Y) \end{array}$$

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- (v) For every sequence of finite maps $I \rightarrow J \rightarrow K$, collections of objects $X_I = \{X_i\}_{i \in I}$, $Y_J = \{Y_j\}_{j \in J}$, $W_K = \{W_k\}_{k \in K}$ and object $Z \in \mathcal{O}$ the following diagram commutes:

make diagram prettier

$$\begin{array}{ccc}
 \prod_{j \in J} \text{Mul}_{\mathcal{O}}(X_{I_j}, Y_j) \times \prod_{k \in K} \text{Mul}_{\mathcal{O}}(Y_j, W_k) \times \text{Mul}_{\mathcal{O}}(W_K, Z) & \xrightarrow{\text{Id} \times \gamma} & \prod_{j \in J} \text{Mul}_{\mathcal{O}}(X_{I_j}, Y_j) \times \text{Mul}_{\mathcal{O}}(Y_J, Z) \\
 \downarrow (\gamma \circ s) \times \text{Id} & & \downarrow \\
 \prod_{k \in K} \text{Mul}_{\mathcal{O}}(X_{I_k}, W_k) \times \text{Mul}_{\mathcal{O}}(W_K, Z) & \xrightarrow{\quad} & \text{Mul}_{\mathcal{O}}(X_I, Z)
 \end{array}$$

Remark 2.23. When we drop the symmetric structure we obtain a definition of a colored operad which we call colored planar operad. In other words we consider the indexing sets I in definition 2.22 to be finite ordered sets and maps between indexing sets have to be order preserving.

Remark 2.24. We note that there is a forgetful functor from the categories of colored operads to the category of small categories. Hence, for every colored operad $\mathcal{O}$ there is an underlying category which we also denote by $\mathcal{O}$. Its objects are the colors of $\mathcal{O}$ and for $X, Y \in \mathcal{O}$ we define the morphisms

$$\text{Hom}(X, Y) := \text{Mul}_{\mathcal{O}}(\{X\}, Y).$$

Composition and identity hold since they hold in $\mathcal{O}$. Thus, a colored operad can be viewed as a category with additional structure. Since we have some sort of collection of multilinear maps, some also refer to colored operads as multicategories.

Examples 2.25. 1. A colored operad with one color is a symmetric operad in **Set** as defined in definition 2.8. Let $\mathcal{O}$ be a colored operad with one color X . We will construct a symmetric operad $\mathcal{P}$ from $\mathcal{O}$. Since $\mathcal{O}$ is a colored operad with only one color, we define the family of objects $\mathcal{P}(n)$ for $n \in \mathbb{N}$ as:

$$\mathcal{P}(n) := \text{Mul}_{\mathcal{O}}(\{X\}_{1 \leq i \leq n}, X).$$

The composition maps from $\mathcal{O}$ for natural number indexing then defines the composition maps of $\mathcal{P}$

$$\gamma : \mathcal{P}(k) \times \mathcal{P}(j_1) \times \cdots \times \mathcal{P}(j_k) \rightarrow \mathcal{P}(j)$$

where $j = \sum_{i=1}^k j_i$ and $\eta : \{*\} \rightarrow \mathcal{P}(1)$ is given by Id_X . Unitality and associativity follow from the corresponding conditions in $\mathcal{O}$. However, in a colored operad (even with one color), the set of morphisms depends not only on n , the cardinality of the indexing set, but also on the specific choice of finite indexing set. Thus, for two sets N_1 and N_2 with the same number of elements, you could potentially have different morphism sets:

$$\text{Mul}_{\mathcal{O}}(\{X_i\}_{i \in N_1}, Y) \quad \text{and} \quad \text{Mul}_{\mathcal{O}}(\{X_i\}_{i \in N_2}, Y).$$

This raises the question of how the structures of a colored operad and an operad reconcile. The reason these two different setups are still equivalent lies in the symmetric group action on the set of morphisms. The permutation of the inputs ensures that any specific labeling of the inputs (as represented by the different finite sets N_1 and N_2) can be transformed into each other via the action of S_n . Hence, although the morphisms in the colored operad are indexed by specific sets, they behave the same way up to this permutation action. Therefore, the group action effectively collapses the dependence on the specific finite sets. So, while the morphism sets in a colored operad are technically different for different finite sets, the permutation symmetry makes these differences irrelevant, aligning the structure with that of an operad.

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2. A colored planar operad with one color is a planar operad in **Set** as defined in definition 2.1. Let $\mathcal{P}$ be a colored planar operad $\mathcal{P}$. We set for $n \in \mathbb{N}$:

$$\mathcal{P}(n) := \text{Mul}_{\mathcal{P}}((X)_{1 \leq i \leq n}, X)$$

We obtain the composition map and operadic unit similarly as in the example for a one colored operad. However, as the indexing maps need to be order preserving we will not obtain a permutation of labels as there is only one bijective order preserving map between finite ordered sets.

Examples 2.26. 1. Let $\mathcal{C}$ be a symmetric monoidal category. By setting for a finite family of objects $\{X_i\}_{i \in I}$ and object Y in $\mathcal{C}$

$$\text{Mul}_{\mathcal{C}}(\{X_i\}_{i \in I}, Y) := \text{Hom}(\otimes_{i \in I} X_i, Y)$$

we can define a colored operad $\mathcal{C}$. The composition is given by

$$\prod_{j \in J} \text{Hom}(\otimes_{i \in I_j} X_i, Y_j) \times \text{Hom}(\otimes_{j \in J} Y_j, Z) \rightarrow \text{Hom}(\otimes_{i \in I} X_i, Z)$$

$$((f_j)_{j \in J}, g) \mapsto g \circ (\otimes_{j \in J} f_j)$$

with $\{\text{Id}_X\}_{X \in \mathcal{C}}$ being the unit and associativity holds since for $(f_j: \otimes_{i \in I_j} X_i \rightarrow Y_j)_{j \in J}$, $(g_k: \otimes_{j \in J_k} Y_j \rightarrow W_k)_{k \in K}$, $h: \otimes_{k \in K} W_k \rightarrow Z$ we have

$$h \circ \bigotimes_{k \in K} \left(g_k \circ \bigotimes_{j \in J_k} f_j \right) = h \circ \bigotimes_{k \in K} g_k \circ \bigotimes_{j \in J} f_j.$$

Further, we can reconstruct the symmetric monoidal structure of $\mathcal{C}$ from this colored operad: Let $F_{X,Y}: \mathcal{C} \rightarrow \mathbf{Set}$ be the functor sending an object Z to $\text{Mul}_{\mathcal{C}}(\{X, Y\}, Z)$ then by definition and from Yoneda lemma follows that F is uniquely (up to isomorphism) represented by $X \otimes Y$. Let $f: X \rightarrow X'$ and $g: Y \rightarrow Y'$ be morphisms in $\mathcal{C}$ and

$$\text{Mul}_{\mathcal{C}}(\{X\}, X') \times \text{Mul}_{\mathcal{C}}(\{Y\}, Y') \times \text{Mul}_{\mathcal{C}}(\{X', Y'\}, X' \otimes Y') \xrightarrow{\sim} \text{Mul}_{\mathcal{C}}(\{X, Y\}, X' \otimes Y')$$

the composition map. Then, we define $f \otimes g := \gamma((f, g, \text{Id}))$. The tensor unit I is the representing object of the functor sending an object Z in $\mathcal{C}$ to $\text{Mul}_{\mathcal{C}}(\emptyset, Z)$. This example is even clearer when we have the functorial definition of a colored operad later on. As it will be a generalization of proposition 2.19 and proposition 2.20 illustrates how to reconstruct the symmetric monoidal category.

2. Analogously, a monoidal category $\mathcal{C}$ is an example of a colored planar operad.

Definition 2.27. Let $\mathcal{O}$ be a colored operad. We define a category $\mathcal{O}^{\otimes}$ in the following way:

- (i) The objects of $\mathcal{O}^{\otimes}$ are finite sequences of colors of $\mathcal{O}$.
- (ii) Let $\{X_i\}_{1 \leq i \leq m}$ and $\{Y_j\}_{1 \leq j \leq n}$ be objects in $\mathcal{O}^{\otimes}$. A morphism

$$\phi: \{X_i\}_{1 \leq i \leq m} \rightarrow \{Y_j\}_{1 \leq j \leq n}$$

is given by a map $\alpha: \langle m \rangle \rightarrow \langle n \rangle$ in Fin_* together with a family of morphisms

$$\{\phi_j \in \text{Mul}_{\mathcal{O}}(\{X_i\}_{i \in \alpha^{-1}(j)}, Y_j)\}_{1 \leq j \leq n}$$

in $\mathcal{O}$.

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- (iii) Composition of morphisms is determined by the composition in $\mathcal{Fin}_*$ and $\mathcal{O}$, i.e. for morphisms $\phi: X \rightarrow Y$ and $\phi': Y \rightarrow Z$ in $\mathcal{O}^\otimes$ with $\alpha: \langle n \rangle \rightarrow \langle m \rangle$ and $\alpha': \langle m \rangle \rightarrow \langle k \rangle$, we have $\bar{\alpha} = \alpha' \circ \alpha$ and

$$\{(\phi' \circ \phi)_l \in \text{Mul}_{\mathcal{O}}(\{X_i\}_{i \in \bar{\alpha}^{-1}(l)}, Z_l)\}_{1 \leq l \leq k}$$

where $(\phi' \circ \phi)_l$ is applying the composition map

$$\gamma: \prod_{j \in \alpha'^{-1}(l)} \text{Mul}_{\mathcal{O}}(\{X_i\}_{i \in \alpha^{-1}(j)}, Y_j) \times \text{Mul}_{\mathcal{O}}(\{Y_i\}_{i \in \alpha'^{-1}(l)}, Z_l) \rightarrow \text{Mul}_{\mathcal{O}}(\{X_i\}_{i \in \bar{\alpha}^{-1}(l)}, Z_l)$$

Remark 2.28. For a planar colored operad $\mathcal{P}$, we can similarly define a category $\mathcal{P}^\otimes$ by requiring $\alpha: [m] \rightarrow [n]$ to be morphism in the simplex category with a distinguished point Δ_* .

Let $\mathcal{O}$ be a colored operad. Then similarly as for the definition 2.15 we have a forgetful functor

$$\pi: \mathcal{O}^\otimes \rightarrow \mathcal{Fin}_*$$

from which we can reconstruct the underlying colored operad $\mathcal{O}$ (up to canonical equivalence): First, note that in general π does not have to be a op-fibration. For an arbitrary morphism $\alpha: \langle n \rangle \rightarrow \langle m \rangle$ and a collection $\{X_i\}_{i \in \langle n \rangle}$ there is no canonical choice for a collection $\{X'_j\}_{j \in \langle m \rangle}$ with morphism $f: \{X_i\}_{i \in \langle n \rangle} \rightarrow \{X'_j\}_{j \in \langle m \rangle}$ such that $\pi(f) = \alpha$ as in the proof of proposition 2.19. However, there is a canonical choice if α is a bijection on $\alpha^{-1}(\langle m \rangle^\circ)$. Then for all $j \in \langle m \rangle$ there is exactly one $i_j \in \langle n \rangle$ with $\alpha(i_j) = j$. Hence, we define $X'_j := X_{i_j}$ and for all $j \in J$ we have $f_j = \text{Id}_{X_{i_j}}$. This defines a morphism $f: \{X_i\}_{i \in I} \rightarrow \{X'_j\}_{j \in J}$.

It is left to show that f is in fact cocartesian. Let $g: \{X_i\}_{i \in I} \rightarrow \{X''_k\}_{k \in \langle l \rangle}$ be a morphism in $\mathcal{O}^\otimes$ and $\omega: \langle m \rangle \rightarrow \langle l \rangle$ a morphism in $\mathcal{Fin}_*$ such that $\pi(g) = \omega \circ \pi(f)$. Due to the fact of $\pi(f)$ being a bijection on the restriction, we observe that ω is determined by $\pi(g)$. It follows that

$$\text{Mul}(\{X'_j\}_{j \in \omega^{-1}(k)}, X''_k) = \text{Mul}(\{X_{a_j}\}_{j \in \omega^{-1}(k)}, X''_k) = \text{Mul}(\{X_i\}_{i \in \pi(g)^{-1}(k)}, X''_k)$$

and we can define for all $k \in \langle l \rangle$ $h_k = g_k$. So, we have constructed a morphism

$$h: \{X'_j\}_{j \in \langle m \rangle} \rightarrow \{X''_k\}_{k \in \langle l \rangle}$$

such that $h \circ f = g$ and $\pi(h) = \omega$. The underlying category of $\mathcal{O}$ is the fiber category $\mathcal{O}_{\langle 1 \rangle}$ as the objects are 1-collections of colors and a morphism from X to Y is defined by $\text{Id}_{\langle 1 \rangle}$ with $\phi \in \text{Mul}_{\mathcal{O}}(\{X\}, Y)$. Furthermore, the equivalence $\mathcal{O}_{\langle n \rangle} \cong \mathcal{O}^n$ holds as the weaker op-fibration condition (M1) property still suffices to admit the proof given in proposition 2.19.

Further, let $\{X_i\}_{i \in \langle n \rangle}$ be a collection of colors, $\vec{X}$ denoting the corresponding object in $\mathcal{O}_{\langle n \rangle}^\otimes$ and $Y \in \mathcal{O}$. The set $\text{Mul}_{\mathcal{O}}(\{X_i\}_{i \in \langle n \rangle}, Y)$ can be identified with the set of morphisms $f: \vec{X} \rightarrow Y$ in $\mathcal{O}^\otimes$ such that $\pi(f)$ being the morphism with $\pi(f)^{-1}(*) = \{*\}$. In addition, the composition law for morphisms in the colored operad $\mathcal{O}$ can be reconstructed from the composition law in $\mathcal{O}^\otimes$.

Remark 2.29. Let $\mathcal{P}$ be a colored planar operad. Then analogously there is also a forgetful functor

$$\pi: \mathcal{P}^\otimes \rightarrow \Delta_*.$$

which satisfies the weaker op-fibration condition (M1) and Segal condition (M2) property.

This gives us an alternative definition of a colored operad. Instead of seeing a colored operad as a categorical structure $\mathcal{O}$ equipped with an extensive notion of morphisms, we can now view colored operads as a category $\mathcal{O}^\otimes$ equipped with a forgetful functor $\pi: \mathcal{O}^\otimes \rightarrow \mathcal{Fin}_*$ satisfying conditions such as $\mathcal{O}_{\langle n \rangle}^\otimes \cong (\mathcal{O}_{\langle 1 \rangle}^\otimes)^n$ and $\mathcal{O} = \mathcal{O}_{\langle 1 \rangle}^\otimes$ inherit the structure of a colored operad. Even though the conditions π has to satisfy seem tedious, the second definition is phrased entirely in the language of category theory and can therefore be generalized to an ∞ -categorical notion of a colored operad.

2. Operads

Definition 2.30. A morphism $f: \langle n \rangle \rightarrow \langle m \rangle$ is called *inert* if for every $i \in \langle m \rangle^\circ$ the inverse image $f^{-1}(i)$ has exactly one element.

Definition 2.31. An ∞ -operad is a functor $p: \mathcal{O}^\otimes \rightarrow \mathcal{N}(\mathcal{F}\text{in}_*)$ between ∞ -categories satisfying the following conditions:

- (i) For every inert morphism $f: \langle n \rangle \rightarrow \langle m \rangle$ in $\mathcal{N}(\mathcal{F}\text{in}_*)$ and object $C \in \mathcal{O}_{\langle m \rangle}^\otimes$ there exists a p -cocartesian morphism $f: C \rightarrow C'$ in $\mathcal{O}^\otimes$ lifting f . In particular f induces a functor $f_!: \mathcal{O}_{\langle n \rangle}^\otimes \rightarrow \mathcal{O}_{\langle m \rangle}^\otimes$.
- (ii) Let $C \in \mathcal{O}_{\langle m \rangle}^\otimes$ and $C' \in \mathcal{O}_{\langle n \rangle}^\otimes$ be objects, $f: \langle m \rangle \rightarrow \langle n \rangle$ be a morphism in $\mathcal{F}\text{in}_*$ and let $\text{Map}_{\mathcal{O}^\otimes}^f(C, C')$ be the union of connected components of $\text{Map}_{\mathcal{O}^\otimes}(C, C')$ which lie over f . Choose p -cocartesian morphisms $C' \rightarrow C'_i$ lying over the inert morphism $\rho^i: \langle n \rangle \rightarrow \langle 1 \rangle$ for $1 \leq i \leq n$. Then the induced map

$$\text{Map}_{\mathcal{O}^\otimes}^f(C, C') \rightarrow \prod_{1 \leq i \leq n} \text{Map}_{\mathcal{O}^\otimes}^{\rho^i \circ f}(C, C'_i)$$

is a homotopy equivalence.

- (iii) For every finite collection of objects $C_1, \dots, C_n \in \mathcal{O}_{\langle 1 \rangle}^\otimes$, there exists an object $C \in \mathcal{O}_{\langle n \rangle}^\otimes$ and a collection of p -cocartesian morphism $C \rightarrow C_i$ covering $\rho^i: \langle n \rangle \rightarrow \langle 1 \rangle$.

3 Braided Operads

Braided operads were first mentioned by Fiedorowicz in his paper [ZF] on the construction of E_n operads. Let A be an operad in the category of compactly generated Hausdorff spaces. The first theorem states:

Theorem 3.1 (Recognition principles for E_n -operads). *A is an E_2 -operad iff $A(k)$ is connected and the collection of universal covering spaces $\{\tilde{A}(k)\}_{k \in \mathbb{N}}$ forms a B_∞ -operad, i.e.:*

- (1) For all $k \in \mathbb{N}$, the space $\tilde{A}(k)$ is contractible.
- (2) The braid group B_k acts freely on $\tilde{A}(k)$ for each $k \in \mathbb{N}$.

In the sketch of the proof, it is stated that "the definition of operads can be reformulated using actions by braid groups in place of actions by symmetric groups" [ZF]. As braided operads are a useful tool, e.g. recognizing E_n -operads, we want to gain a deeper understanding of braided operads. Further, we have already seen that colored operads correspond to op-fibrations over finite pointed sets Fin_* as well as colored planar operads correspond to op-fibrations over Δ_* . We are hoping to find a suitable category such that colored braided operads correspond to op-fibrations over this category.

It might be interesting to try to define braided operads in braided monoidal categories rather than symmetric monoidal categories potentially yielding interesting examples. However, it is not straight forward how to do this as shuffling objects would not be a canonical isomorphism. So for now, we will stay in a symmetric monoidal category $\mathcal{C}$.

Definition 3.2. A *braided operad* $\mathcal{O}$ in $\mathcal{C}$ is a planar operad as defined in definition 2.1 with an action of B_n on $\mathcal{O}(n)$ for all $n \in \mathbb{N}$ together with the following equivariance condition: For $\sigma \in B_k$, let $\sigma(j_1, \dots, j_k) \in B_j$ denote the element that σ -braids k blocks with each block having j_l strands for $1 \leq l \leq k$. Then the following diagram commutes:

$$\begin{array}{ccc} \mathcal{O}(k) \otimes \mathcal{O}(j_1) \otimes \dots \otimes \mathcal{O}(j_k) & \xrightarrow{\sigma \otimes \bar{\sigma}^{-1}} & \mathcal{O}(k) \otimes \mathcal{O}(j_{\bar{\sigma}(1)}) \otimes \dots \otimes \mathcal{O}(j_{\bar{\sigma}(k)}) \\ \gamma \downarrow & & \downarrow \gamma \\ \mathcal{O}(j) & \xrightarrow{\sigma(j_1 \dots j_k)} & \mathcal{O}(j), \end{array}$$

where $\bar{\sigma}$ denotes the underlying permutation of σ . Furthermore, let $\tau_l \in B_{j_l}$ for $1 \leq l \leq k$ and $\tau_1 \oplus \dots \oplus \tau_k$ denotes the sum of braidings on k blocks with each block having j_l strands. Then the following diagram commutes

$$\begin{array}{ccc} \mathcal{O}(k) \otimes \mathcal{O}(j_1) \otimes \dots \otimes \mathcal{O}(j_k) & \xrightarrow{\text{Id} \otimes \tau_1 \otimes \dots \otimes \tau_k} & \mathcal{O}(k) \otimes \mathcal{O}(j_1) \otimes \dots \otimes \mathcal{O}(j_k) \\ \gamma \downarrow & & \downarrow \gamma \\ \mathcal{O}(j) & \xrightarrow{\tau_1 \oplus \dots \oplus \tau_k} & \mathcal{O}(j) \end{array}$$

Remark 3.3. Similarly to symmetric operads, we can define algebras and modules over braided operads by changing the equivariance condition:

Let $\mathcal{O}$ be a braided operad and A an algebra over the underlying planar operad of $\mathcal{O}$. For all $j \in \mathbb{N}$, $\sigma \in B_j$ and $\bar{\sigma} \in S_j$ being the underlying permutation of σ :

$$\begin{array}{ccc} \mathcal{O}(j) \otimes A^j & \xrightarrow{\sigma \otimes \bar{\sigma}^{-1}} & \mathcal{O}(j) \otimes A^j \\ & \searrow \theta & \swarrow \theta \\ & A & \end{array}$$

Then, we call A an algebra over the braided operad $\mathcal{O}$.

Further, let A be an algebra over the braided operad $\mathcal{O}$ and M an object in $\mathcal{C}$ with

$$\omega: \mathcal{O}(j) \otimes A^{j-1} \otimes M \rightarrow M$$

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satisfying associativity and unitality conditions from definition 2.10. Then we call M an A -module if the following equivariance condition holds for all $j \in \mathbb{N}$, $\sigma \in B_{j-1} \subset B_j$ and $\bar{\sigma} \in S_{j-1}$ being the underlying permutation of σ :

$$\begin{array}{ccc} O(j) \otimes A^{j-1} \otimes M & \xrightarrow{\sigma \otimes \bar{\sigma}^{-1} \otimes \text{Id}} & O(j) \otimes A^{j-1} \otimes M \\ & \searrow \omega & \swarrow \omega \\ & M & \end{array}$$

- Example 3.4.** 1. Again, let us consider the operad in **Set** defined by the singleton set in each degree $n \in \mathbb{N}$. The braid group acts trivially. When we consider algebras over this operad we will again obtain commutative algebras.
2. We have already encountered an example of a planar operad in example 2.6 which can be lifted to a braided operad—the braid operad $\mathcal{B}$. The braid group B_n acts by composition. This group action is equivariant.

3.1 Action Operad

For this section, we remain in the symmetric monoidal category of sets equipped with the cartesian symmetric monoidal structure. We will introduce an action operad that acts on planar operads and this relates to braided operads following [DY]. An idea will be that the theory on braided operads can be extended in a sense to group action operads.

Definition 3.5. An *action operad* is a tuple

$$(\{G(n)\}_{n \in \mathbb{N}}, \gamma^G, 1^G, \omega)$$

satisfying the following conditions:

- (i) For all $n \in \mathbb{N}$, $G(n)$ is a group with unit Id_n .
- (ii) There is a group homomorphism $\omega: G(n) \rightarrow S_n$ for all n which we call augmentation. For each $g \in G(n)$ we call $\omega(g)$ the underlying permutation of g .
- (iii) The tuple $(\{G(n)\}_{n \in \mathbb{N}}, \gamma^G, 1^G)$ is a one-colored planar operad in **Set** which we denote by G . It fulfills the following conditions:

- (a) The augmentation

$$\omega: G \rightarrow \mathcal{S}$$

is a morphism of planar operads in **Set** where $\mathcal{S}$ denotes the symmetry operad from examples 2.12.

- (b) For $n \geq 1$, let $\sigma, \rho \in G(n)$ and for $k_j \in \mathbb{N}$, $1 \leq j \leq n$ let $\tau_j, \tau'_j \in G(k_j)$. Then the following equation holds in $G(k)$ for $\sum_{1 \leq j \leq n} k_j = k$:

$$\gamma^G(\sigma \rho, \tau_1 \tau'_1, \dots, \tau_n \tau'_n) = \gamma^G(\sigma, \tau_{\bar{\rho}^{-1}(1)}, \dots, \tau_{\bar{\rho}^{-1}(n)}) \cdot \gamma^G(\rho, \tau'_1, \dots, \tau'_n)$$

where $\bar{\rho} = \omega(\rho) \in S_n$.

We call the condition (iii)(b) the action operad equation.

- Examples 3.6.** 1. The *planar group operad* P is the action operad where $P(n) = \{*\}$ is the trivial group for all $n \in \mathbb{N}$.
2. The symmetry operad $\mathcal{S}$ from examples 2.12 equipped with the identity augmentation forms an action operad.
3. Another example which we have already seen is the braid operad which we equip with the projection to the symmetric group as the augmentation.

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Proposition 3.7. *The braid operad $\mathcal{B}$ from example 2.6 equipped with the projection*

$$\pi: \mathcal{B} \rightarrow \mathcal{S}$$

forms an action operad

$$B = (\{B_n\}_{n \in \mathbb{N}}, \gamma, \text{Id}_1 \in \mathcal{B}_1, \pi)$$

Proof. We have already seen that $\mathcal{B}$ forms a one-colored planar operad in **Set**. Hence, it remains to show that π is a morphism of planar operads. We note that $\pi(\text{Id}_1)$ is the trivial permutation and π preserves the operadic composition as it preserves the group multiplication. The action operad equation follows from remark 2.5. $\square$

3.2 Universal Covering of E_2

In this section, we want to establish an operadic structure on the sequence of levelwise universal coverings of E_2 . We recall the operad E_2 from Example 2.12 which we introduced as the little 2-disks operad. So in each configuration space $E_2(n)$ we have n disks embedded into a large disk by translation and resizing which are pairwise disjoint. We use topological terms when talking about E_2 what we really mean is applying the term level-wise. We will see that the universal coverings of $E_2(n)$ for all $n \in \mathbb{N}$ can be equipped with an operadic structure as well as a braid action. Hence, we obtain another example of a braided operad.

The operadic structure of E_2 requires each configuration to have labels as they are necessary to define the composition maps. However, the images of the disk embeddings of configurations that only differ in their labels are identical. It will be useful to identify configurations that only differ in their labels. Therefore, consider the quotient under the symmetric group action for all $n \in \mathbb{N}$:

$$E_2(n) \rightarrow E_2(n)/S_n$$

We denote the family of these quotient spaces by E_2/S . Note that this is not an operad any longer as the composition map requires labels.

The following lemma will be essential to establish a braid group action on the universal covering $\widetilde{E_2(n)}$.

Lemma 3.8. *The following statements hold for every configuration $E_2(n)$ for all $n \in \mathbb{N}$:*

1. *The fundamental group of $E_2(n)$ is the pure braid group PB_n .*
2. *The fundamental group of $E_2(n)/S_n$ is the braid group B_n .*

We will not prove the lemma here. A proof by Fox and Neuwirth can be found in [FN62]. However, we will give a sketch of why this is the case at the end of the section.

Before we get started, we need to define what it means to be a universal covering of a topological operad.

Definition 3.9. Let $\mathcal{P}$ be a topological operad. A covering operad $\widetilde{\mathcal{P}}$ is a topological operad such that there is a morphism of operads $c: \widetilde{\mathcal{P}} \rightarrow \mathcal{P}$ where $c_n: \widetilde{\mathcal{P}}(n) \rightarrow \mathcal{P}(n)$ is a covering of $\mathcal{P}(n)$ for all $n \geq 1$.

Definition 3.10. Let $\mathcal{P}$ be a topological operad. A universal covering operad $\widetilde{\mathcal{P}}$ is a covering operad that is universal in every degree $n \geq 1$.

To be able to talk about universal covers and fundamental groups we need to choose a basepoint. So formally, for any choice of a basepoint on each level n , permuting its labels will give us a different point in $E_2(n)$. Hence, if we consider a loop at a basepoint we are required to end the loop with the same labeling of the configuration of the basepoint. That is the reason why when considering the fundamental group of $E_2(n)$ over some basepoint we only obtain the pure braid group on n strands by Lemma 3.8. In this section we want to establish a universal covering of E_2 which has the full braid action. In the next step, we will first define a universal covering of E_2 and E_2/S for each level n . Then, we will see that these spaces are homeomorphic on

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3. Braided Operads

each level. This allows us to transfer the braid action on the universal covering of E_2/S to the universal covering of E_2 .

In this part, we will work a lot with $E_2^\otimes$ which is Definition 2.27 applied to E_2 . This gives us a nice way to talk about configurations and paths between them. Hence, let us recall the definition of $E_2^\otimes$:

Remark 3.11. The objects of $E_2^\otimes$ are the finite pointed sets $\langle n \rangle$ for all $n \in \mathbb{N}$. A morphism $\langle n \rangle \rightarrow \langle m \rangle$ in $E_2^\otimes$ consists of a morphism $\alpha: \langle n \rangle \rightarrow \langle m \rangle$ in $\mathcal{Fin}_*$ and a collection of configurations

$$\{c_j \in E_2(|\alpha^{-1}(j)|)\}_{1 \leq j \leq m}$$

where each configuration is labeled by $\alpha^{-1}(j)$.

We denote the trivial configuration with label x by $\text{Id}^x \in E_2(1)$. For $\sigma \in S_n$ we refer to the corresponding automorphism of $\langle n \rangle$ with configurations $(\text{Id}^1, \dots, \text{Id}^n)$ also by σ . Further, recall the composition in $E_2^\otimes$. Let

$$\alpha: \langle n \rangle \rightarrow \langle m \rangle \text{ with configurations } \{c_j\}_{1 \leq j \leq m}$$

and

$$\alpha': \langle m \rangle \rightarrow \langle l \rangle \text{ with configurations } \{c'_k\}_{1 \leq k \leq l}.$$

Then, the configuration c'_k is labeled by $\alpha'^{-1}(k) = \{j_1^{(k)}, \dots, j_p^{(k)}\}$. So, with the composition map γ we obtain a configuration

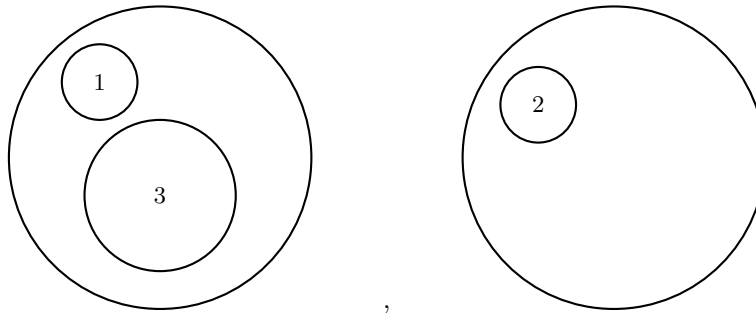
$$\gamma(c'_k, c_{j_1}, \dots, c_{j_p})$$

labeled by $(\alpha' \circ \alpha)^{-1}(k)$ for all $1 \leq k \leq l$ giving us the well-defined composition $\alpha' \circ \alpha$.

Example 3.12. An example of a morphism in $E_2^\otimes$ is the following:

$$\begin{aligned} \alpha: \langle 3 \rangle &\rightarrow \langle 2 \rangle \\ 1 &\mapsto 1 \\ 2 &\mapsto 2 \\ 3 &\mapsto 1 \end{aligned}$$

together with the configurations:



The category $E_2^\otimes$ will be an essential ingredient to establish colored braided operads.

So, in $E_2^\otimes$ we identify $f, g: \langle n \rangle \rightarrow \langle m \rangle$ if there exists a commuting diagram in $E_2^\otimes$

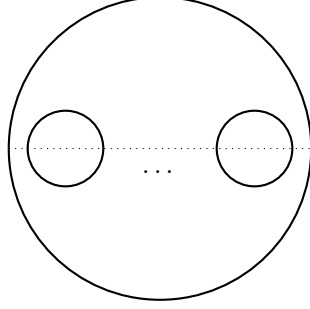
$$\begin{array}{ccc} \langle n \rangle & \xrightarrow{f} & \langle m \rangle \\ \downarrow \sigma & & \downarrow \tau \\ \langle n \rangle & \xrightarrow{g} & \langle m \rangle \end{array}$$

where $\sigma \in S_n$ and $\tau \in S_m$.

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Construction 3.13. We begin the construction of a universal covering of E_2/S by choosing a basepoint in each configuration space. So, for all $n \geq 1$ we choose the basepoint $\bar{b}_n \in E_2(n)/S_n$ to be the following configuration:



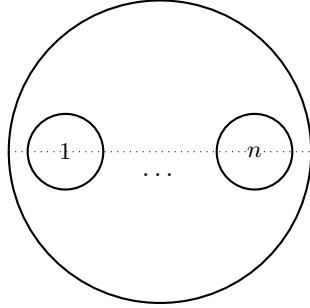
Note that since $E_2(n)$ is connected and simply connected, the quotient $E_2(n)/S_n$ is connected. Hence, it has a universal covering $\widetilde{E_2(n)/S_n}$ which we define:

$$\widetilde{E_2(n)/S_n} := \{\gamma: [0, 1] \rightarrow E_2(n)/S_n \mid \gamma(0) = \bar{b}_n\} / \text{homotopy with fixed endpoints}$$

with projection map:

$$\begin{aligned} p: \widetilde{E_2(n)/S_n} &\rightarrow E_2(n)/S_n \\ \gamma &\mapsto \gamma(1) \end{aligned}$$

The topology of the covering and the fact that it defines a universal covering of $E_2(n)/S_n$ is shown in Section 1.3 in [Hat02]. Analogously, there is a universal covering of E_2 , when we choose the basepoint $b_n \in E_2(n)$:



and define its universal covering analogously:

$$\widetilde{E_2(n)} := \{\gamma: [0, 1] \rightarrow E_2(n) \mid \gamma(0) = b_n\} / \text{homotopy with fixed endpoints}$$

with projection map:

$$\begin{aligned} p: \widetilde{E_2(n)} &\rightarrow E_2(n) \\ \gamma &\mapsto \gamma(1) \end{aligned}$$

There is a homeomorphism for each $n \in \mathbb{N}$:

$$\begin{aligned} \widetilde{E_2(n)} &\cong \widetilde{E_2/S(n)} \\ (\gamma: b_n \rightsquigarrow c) &\mapsto (\bar{\gamma}: \bar{b}_n \rightsquigarrow \bar{c}) \\ (\gamma: b_n \rightsquigarrow c') &\mapsto (\bar{\gamma}: \bar{b}_n \rightsquigarrow \bar{c}) \end{aligned}$$

3. Braided Operads

where we forget labels in one direction and in the reverse direction we label the basepoint $\bar{b}_n$ from left to right. The labeling of c' is then induced by the path γ . Clearly this is continuous in both directions and inverse to one another. This shows that topologically the path γ does not need the labels which is summarized in the following notation. For a path $\gamma: \bar{b}_n \rightsquigarrow c$ between configurations $\bar{b}_n, c \in \widetilde{E_2(n)}/S_n$ we write the following diagram in $E_2^\otimes$ for some $\sigma \in S_n$:

$$\begin{array}{ccc} \langle n \rangle & \xrightarrow{c} & \langle 1 \rangle \\ \downarrow \sigma & & \uparrow \gamma \\ \langle n \rangle & \xrightarrow{b_n} & \langle 1 \rangle \end{array}$$

Note, that the dashed arrow is not a morphism in $E_2^\otimes$ but a path of configurations in $E_2(n)$. Hence, the following diagram would be more accurate, but our notation will be more convenient in this part.

$$\begin{array}{ccc} \langle n \rangle & \xrightarrow{\sigma} & \langle n \rangle \\ & \searrow c & \nearrow b_n \\ & \langle 1 \rangle & \end{array}$$

Essentially, $\sigma \in S_n$ fixes the labels such that we can consider γ as a path in $\widetilde{E}_2$.

We will use this representation to define an operadic structure on $\widetilde{E}_2/S$ such that the covering is a morphism of operads and then also use it to define a braid action on the universal covering. First, we notice a specific translation between specific configurations that are unique up to homotopy equivalence.

Remark 3.14 (E_1 translation). First, we observe that there is an embedding of E_1 into E_2 along the horizontal axis.

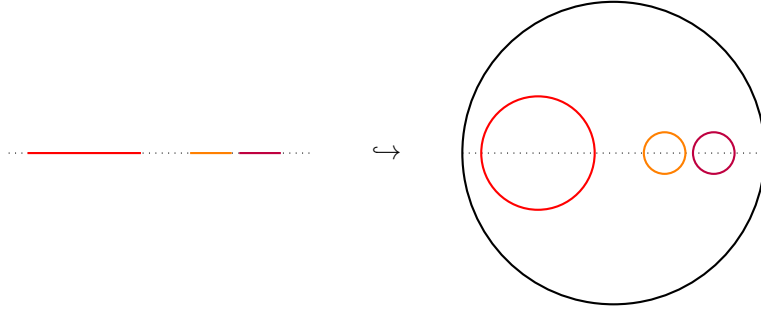


Figure 1: example of a configuration of E_1 being embedded in E_2 along the horizontal line

Further for any $n \in \mathbb{N}$, two configurations $c_1, c_2 \in E_2(n)$ that lie in the image of the embedding, i.e. their middle points lie on the horizontal axis, have a path $\eta: c_1 \rightsquigarrow c_2$ corresponding to a path between the preimages in $E_1(n)$. This path is unique up to homotopy equivalence. This follows from the fact that the ordered configuration space of the real line

$$\text{Conf}_n(\mathbb{R}) = \{(x_1, x_2, \dots, x_n) \in \mathbb{R}^n \mid x_1 < x_2 < \dots < x_n\}$$

is contractible. We can define the following map:

$$(x_1, \dots, x_n) \mapsto \left(\frac{x_2 - x_1}{x_n - x_1}, \frac{x_3 - x_1}{x_n - x_1}, \dots, \frac{x_{n-1} - x_1}{x_n - x_1} \right),$$

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which maps $\text{Conf}_n(\mathbb{R})$ homeomorphically onto the open standard simplex:

$$\text{int}(\Delta^{n-2}) = \{(t_1, \dots, t_{n-2}) \in (0, 1)^{n-2} \mid 0 < t_1 < \dots < t_{n-2} < 1\}.$$

Since the open simplex $\text{int}(\Delta^{n-2})$ is contractible then so is $\text{Conf}_n(\mathbb{R})$. We refer to such a unique path between two configurations in the image of the embedding by E_1 -translation. Note that this discussion holds for any embedding of E_1 into E_2 .

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Proposition 3.15. *There is an operadic structure on $\widetilde{E}_2$ such that the covering p is a morphism of operads.*

Proof. Let

$$\gamma: E_2(k) \times E_2(j_1) \times \dots \times E_2(j_k) \rightarrow E_2(j)$$

be a composition map in E_2 . Then, we define a composition map in $\widetilde{E}_2$

$$\tilde{\gamma}: \widetilde{E}_2(k) \times \widetilde{E}_2(j_1) \times \dots \times \widetilde{E}_2(j_k) \rightarrow \widetilde{E}_2(j),$$

which we construct as follows: Let $(\bar{\alpha}, \alpha_1, \dots, \alpha_k) \in \widetilde{E}_2(k) \times \widetilde{E}_2(j_1) \times \dots \times \widetilde{E}_2(j_k)$ with

$$\alpha_l: b_{j_l} \rightsquigarrow c_l$$

for $1 \leq l \leq k$ and

$$\bar{\alpha}: b_k \rightsquigarrow c.$$

Then

$$\bar{\alpha} \circ (\alpha_1, \dots, \alpha_k)$$

denotes the path from $\gamma((b_k, b_{j_1}, \dots, b_{j_k}))$ to $\gamma(c, c_1, \dots, c_k)$ where we first apply

$$\alpha_l: b_{j_l} \rightsquigarrow c_l$$

on the configuration in the l -th disk of b_k for all $1 \leq l \leq k$ and then apply

$$\bar{\alpha}: b_k \rightsquigarrow c$$

on the configuration b_k .

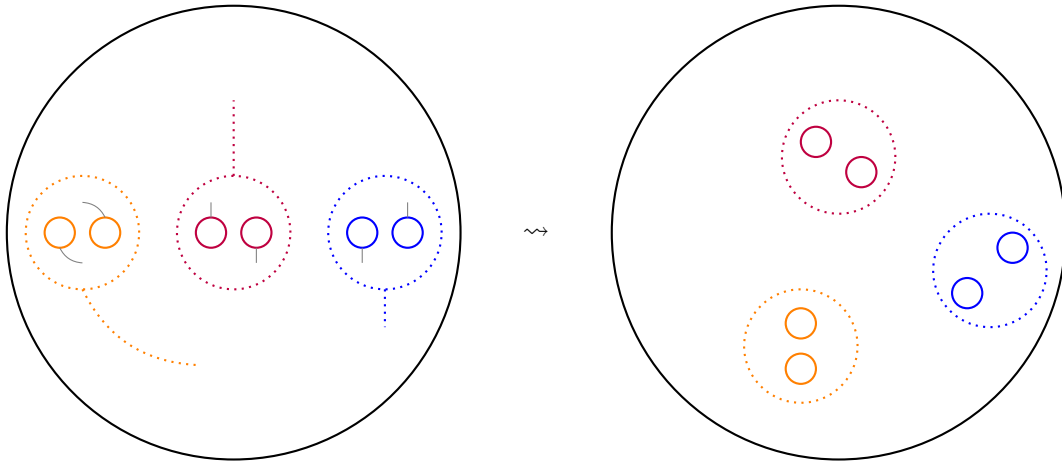


Figure 2: example of $\bar{\alpha} \circ (\alpha_1, \dots, \alpha_3)$ for $\bar{\alpha} \in \widetilde{E}_2(3)$ and $\alpha_1, \alpha_2, \alpha_3 \in \widetilde{E}_2(2)$

Further, let τ be the unique E_1 translation from b_j to $\gamma((b_k, b_{j_1}, \dots, b_{j_k}))$ as discussed in Remark 3.14 that resizes and translates the disks along the horizontal axis. We obtain the following diagram:

3. Braided Operads

$$\begin{array}{ccc}
\langle j \rangle & \xrightarrow{\gamma(c, c_1, \dots, c_k)} & \langle 1 \rangle \\
S_j \Downarrow & & \uparrow \bar{\alpha} \circ (\alpha_1, \dots, \alpha_k) \\
\langle j \rangle & \xrightarrow{\gamma(b_k, b_{j_1}, \dots, b_{j_k})} & \langle 1 \rangle \\
S_j \Downarrow & & \uparrow \tau \\
\langle j \rangle & \xrightarrow{b_j} & \langle 1 \rangle
\end{array}$$

Therefore, we can set

$$\tilde{\gamma}((\bar{\alpha}, \alpha_1, \dots, \alpha_k)) := \bar{\alpha} \circ (\alpha_1, \dots, \alpha_k) \circ \tau$$

We observe that this composition map is well-defined, since the path τ is unique up to homotopy. To see that $\tilde{\gamma}$ is associative we need to have a look at the definition. Note, that in the definition $\tau \in E_1(j)$ is a path that resizes and translates the disks along the horizontal line. Hence, it no braid crossing. Further, we know that composition of paths is associative up to homotopy. So, for

$$(\alpha, (\alpha_s)_{1 \leq s \leq k}, (\alpha_r)_{1 \leq r \leq j}) \in E_2(k) \times \prod_{s=1}^k E_2(j_s) \times \prod_{r=1}^j E_2(i_r)$$

we have a homotopic equivalence:

$$(\alpha \circ (\alpha_s)_{1 \leq s \leq k} \circ \tau^{(k)}) \circ (\alpha'_r)_{1 \leq r \leq j} \circ \tau^{(j)} \simeq \alpha \circ (\alpha_s \circ (\alpha_{g_s+q})_{1 \leq q \leq j_s} \circ \tau^{(j_s)})_{1 \leq s \leq k} \circ \tau^{(k)}$$

Furthermore, we define a operadic unit

$$\eta: * \rightarrow \widetilde{E_2(1)}$$

to be the trivial path of the trivial basepoint $\text{Id}^1 \in E_2(1)$. Clearly, this unit fulfills the unitality conditions.

Finally, we need to show that the covering map $p: \widetilde{E_2} \rightarrow E_2$ is a morphism of operads. For this, we need to convince ourselves that the following diagram commutes for all $n \geq 1$:

$$\begin{array}{ccc}
\widetilde{E_2(k)} \times \widetilde{E_2(j_1)} \times \dots \times \widetilde{E_2(j_k)} & \xrightarrow{p} & E_2(k) \times E_2(j_1) \times \dots \times E_2(j_k) \\
\downarrow \tilde{\gamma} & & \downarrow \gamma \\
\widetilde{E_2(j)} & \xrightarrow{p} & E_2(j)
\end{array}$$

This is indeed the case, by definition of $\tilde{\gamma}$ we have:

$$p \circ \tilde{\gamma}((\bar{\alpha}, \alpha_1, \dots, \alpha_k)) = \gamma(c, c_1, \dots, c_k)$$

for $\alpha_l: b_{j_l} \rightarrow c_l$ for $1 \leq l \leq k$ and $\bar{\alpha}: b_k \rightarrow c$. Clearly, the unitality condition is also met by definition:

$$\begin{array}{ccc}
& * & \\
\tilde{\eta} \swarrow & & \searrow \eta \\
\widetilde{E_2(1)} & \xrightarrow{p} & E_2(1)
\end{array}$$

This concludes the proof that $\widetilde{E_2}$ admits an operadic structure and the covering map is a morphism of operads. $\square$

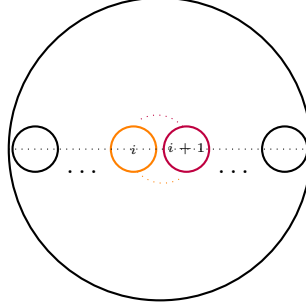
Remark 3.16. This construction of an operadic structure on the universal covering of a topological operad can be given in a more general setting, which we did not require at this point. But in [AB, Lemma 4.1.12] you can find a more general statement on how to construct an operadic structure on the collection of level-wise universal coverings of a sufficiently connected topological operad as long as there is a suboperad of E_1 into the topological operad.

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Construction 3.17 (Braid Action on $\widetilde{E_2/S}$). We want to define a braid action on the universal covering $\widetilde{E_2}$. As we have seen previously, the universal covering of E_2/S is level-wise homeomorphic to the universal covering of E_2 . As usual, there is a group action of the fundamental group of $E_2(n)/S(n)$ on its universal covering. By Lemma ?? the fundamental group of $E_2(n)$ is the n -th pure braid group PB_n . But since we forgot the labels when moving to $\widetilde{E_2(n)/S(n)}$, we obtain now the full n -th braid group B_n . Since, there is a homeomorphism $\widetilde{E_2/S} \cong \widetilde{E_2}$, we can lift the braid group action to $\widetilde{E_2}$. For a detailed proof, we refer to the paper of Fox and Neuwirth [FN62]. However, we give a short overview of the construction here. We recall the generators of B_n for all $1 \leq i < n, n \in \mathbb{N}$:

$$\sigma_i = \begin{array}{c} \left| \quad \quad \quad \begin{array}{c} \text{---} \text{---} \text{---} \\ \diagup \quad \diagdown \\ \text{---} \text{---} \text{---} \\ \diagdown \quad \diagup \\ \text{---} \text{---} \text{---} \end{array} \quad \left| \right. \\ \dots \quad \quad \quad i \quad i+1 \quad \quad \dots \end{array}$$

We identify the generators with the homotopy class of the half twist of the i -th and $(i+1)$ -th disk in $\widetilde{E_2/S}$ which we will also denote by σ_i from now on:



Then, we obtain a braid action on $\widetilde{E_2}$. Let $\alpha: b_n \rightarrow c$ be in $\widetilde{E_2/S}$ which is represented by the following diagram in $E_2^\otimes$ for some $\tau \in S_n$:

$$\begin{array}{ccc} \langle n \rangle & \xrightarrow{c} & \langle 1 \rangle \\ \tau \downarrow & & \uparrow \alpha \\ \langle n \rangle & \xrightarrow{b_n} & \langle 1 \rangle \end{array}$$

and let $\sigma \in B_n$, then we set:

$$\sigma.\alpha := \alpha \circ \sigma^{-1}$$

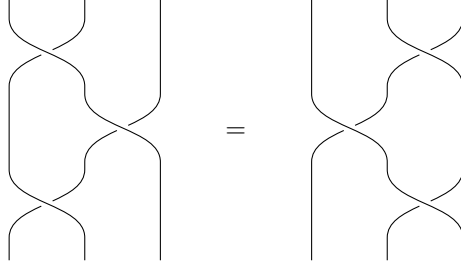
Concatenating these paths results in the following commuting diagram, where $\bar{\sigma} \in S_n$ refers to the underlying permutation of σ .

$$\begin{array}{ccc} \langle n \rangle & \xrightarrow{c} & \langle 1 \rangle \\ \tau \downarrow & & \uparrow \alpha \\ \langle n \rangle & \xrightarrow{b_n} & \langle 1 \rangle \\ \bar{\sigma} \downarrow & & \uparrow \sigma^{-1} \\ \langle n \rangle & \xrightarrow{b_n} & \langle 1 \rangle \end{array}$$

What is left to show is that this braid action is equivariant with respect to the operadic structure on $\widetilde{E_2}$. Note that by definition of $\tilde{\gamma}$ we move disks in a large disk D from the basepoint configuration to another configuration once and we do not touch the configuration of D again afterwards but only move D as a part of another configuration. Hence, in that sense the order

3. Braided Operads

in which we apply paths to a configuration does not matter from a homotopy perspective. This fact is equivalent to the fact that we can pull inside braids through outside braids, i.e. the second relation of braids in Remark 2.4:



This means that for all $\sigma \in B_k$ we have the following commutative diagram:

$$\begin{array}{ccc} \widetilde{E}_2(k) \times \widetilde{E}_2(j_1) \times \cdots \times \widetilde{E}_2(j_k) & \xrightarrow{\sigma \times \bar{\sigma}} & \widetilde{E}_2(k) \times \widetilde{E}_2(j_{\bar{\sigma}(1)}) \times \cdots \times \widetilde{E}_2(j_{\bar{\sigma}(k)}) \\ \downarrow \widetilde{\gamma} & & \downarrow \widetilde{\gamma} \\ \widetilde{E}_2(j) & \xrightarrow{\sigma(j_1, \dots, j_k)} & \widetilde{E}_2(j) \end{array}$$

and for $\tau_i \in B_{j_i}$ with $1 \leq i \leq k$, we have the following commutative diagram:

$$\begin{array}{ccc} \widetilde{E}_2(k) \times \widetilde{E}_2(j_1) \times \cdots \times \widetilde{E}_2(j_k) & \xrightarrow{(\text{Id}, \tau_1, \dots, \tau_k)} & \widetilde{E}_2(k) \times \widetilde{E}_2(j_1) \times \cdots \times \widetilde{E}_2(j_k) \\ \downarrow \widetilde{\gamma} & & \downarrow \widetilde{\gamma} \\ \widetilde{E}_2(j) & \xrightarrow{\tau_1 \oplus \cdots \oplus \tau_k} & \widetilde{E}_2(j) \end{array}$$

Hence, the braid group action on $\widetilde{E}_2$ respects the operadic structure.

4 Colored Braided Operads

In this section, we will define colored braided operads and will aim to find a functor that encodes all the necessary data to give an alternative definition so the notion can eventually be lifted to a ∞ -categorical context.

4.1 Colored Braided Operads

Definition 4.1. Let I be a finite set. For a bijection of finite sets $\alpha: I \cong \langle n \rangle^\circ$ we call a configuration $c \in E_2(n)$ I -labeled.

Definition 4.2. Let I be a finite set with $|I| = n$, c and c' I -labeled configurations corresponding to the same bijection $I \cong \langle n \rangle^\circ$. A path from c to c' is a path between the configurations in $E_2(n)$, i.e. a continuous map

$$f: [0, 1] \rightarrow E_2(n)$$

with $f(0) = c$ and $f(1) = c'$.

Definition 4.3. Let $\mathcal{Br}$ be the following **Grpd**-enriched category:

- (i) The objects of $\mathcal{Br}$ are finite sets.
- (ii) A morphism $\alpha: I \rightarrow J$ in $\mathcal{Br}$ consists of a map of finite sets $\alpha': I \rightarrow J$ and a collection of configurations

$$\{\tilde{\alpha}_j \in E_2(|\alpha'^{-1}(j)|)\}_{j \in J}$$

where each configuration $\tilde{\alpha}_j$ is $\alpha'^{-1}(j)$ -labeled.

- (a) For morphisms

$$\alpha: I \rightarrow J \text{ and } \beta: J \rightarrow K$$

the composition $\beta \circ \alpha$ consists of the map of finite sets

$$\beta' \circ \alpha': I \rightarrow K$$

and for all $k \in K$ with

$$\beta'^{-1}(k) = \{j_1, \dots, j_p\}$$

the $(\beta' \circ \alpha')^{-1}(k)$ -labeled configuration

$$\widetilde{(\beta \circ \alpha)_k} := \gamma(\tilde{\beta}_k, \tilde{\alpha}_{j_1}, \dots, \tilde{\alpha}_{j_p}),$$

where γ is the composition map in E_2 .

- (b) The identity morphism Id_I is given by the identity on finite set I and the collection of trivial configurations $\{\text{Id}^i \in E_2(1)\}_{i \in I}$.
- (iii) For morphisms $\alpha, \beta: I \rightarrow J$ in $\mathcal{Br}$ with $\alpha' = \beta'$ as maps of finite sets. A 2-morphism in $\mathcal{Br}$ consists of a family of homotopy classes of paths

$$\left\{ \eta_j: \tilde{\alpha}_i \Rightarrow \tilde{\beta}_j \right\}_{j \in J}.$$

This defines a 2-category.

We are now ready to define a colored braided operad:

Definition 4.4. A *colored braided operad* $\mathcal{B}$ consists of the following data:

- (i) A collection $\{X, Y, Z, \dots\}$ of colors.

4. Colored Braided Operads

- (ii) For every finite set I , I -labeled configuration $\tilde{I}$, every collection of colors $\{X_i\}_{i \in I}$, and a color Y in $\mathcal{B}$, a set

$$\text{Mul}_{\mathcal{B}}^{\tilde{I}}(\{X_i\}_{i \in I}, Y)$$

called the set of morphisms from $\{X_i\}_{i \in I}$ to Y .

- (iii) For $\alpha: I \rightarrow J$ being a morphism in $\mathcal{B}\mathbf{r}$ with its collection of configurations labeled by the preimages of α :

$$\tilde{\alpha} = (\tilde{I}_j)_{j \in J}$$

where I_j denotes the fiber over j for all $j \in J$, a 2-morphism η :

$$\begin{array}{ccc} I & \xrightarrow{\tilde{\alpha}} & J \\ & \searrow \tilde{I} & \swarrow \tilde{J} \\ & \{1\} & \end{array} \quad \begin{array}{c} \Downarrow \eta \end{array}$$

a collections of colors $\{X_i\}_{i \in I}$, $\{Y_j\}_{j \in J}$ and color Z , a composition map:

$$\gamma: \prod_{j \in J} \text{Mul}_{\mathcal{B}}^{\tilde{I}_j}(\{X_i\}_{i \in I_j}, Y_j) \times \text{Mul}_{\mathcal{B}}^{\tilde{J}}(\{Y_j\}_{j \in J}, Z) \rightarrow \text{Mul}_{\mathcal{B}}^{\tilde{I}}(\{X_i\}_{i \in I}, Z)$$

- (iv) For any I -labeled configuration c with $|I| = 1$, a collection of morphisms

$$\{\text{Id}_X \in \text{Mul}_{\mathcal{B}}^c(\{X\}, X)\}_{X \in \mathcal{B}}$$

called the identity morphisms.

such that the following properties hold:

1. The identity morphisms are a left and right unit with respect to the composition map γ , i.e. the following diagrams commute:

$$\begin{array}{ccc} \text{Mul}_{\mathcal{B}}^{\tilde{I}}(\{X_i\}_{i \in I}, Y) & \xrightarrow{\cong} & \text{Mul}_{\mathcal{B}}^{\tilde{I}}(\{X_i\}_{i \in I}, Y) \times \{\text{Id}_Y\} \\ \downarrow \text{Id} & & \downarrow \\ \text{Mul}_{\mathcal{B}}^{\tilde{I}}(\{X_i\}_{i \in I}, Y) & \xleftarrow{\gamma} & \text{Mul}_{\mathcal{B}}^{\tilde{I}}(\{X_i\}_{i \in I}, Y) \times \text{Mul}_{\mathcal{B}}^c((Y), Y) \end{array}$$

$$\begin{array}{ccc} \text{Mul}_{\mathcal{B}}^{\tilde{I}}(X_I, Y) & \xrightarrow{\cong} & (\prod_{i \in I} \{\text{Id}_{X_i}\}) \times \text{Mul}_{\mathcal{B}}^{\tilde{I}}(X_I, Y) \\ \downarrow \text{Id} & & \downarrow \\ \text{Mul}_{\mathcal{B}}^{\tilde{I}}(X_I, Y) & \xleftarrow{\gamma} & (\prod_{i \in I} \text{Mul}_{\mathcal{B}}^c((X_i), X_i)) \times \text{Mul}_{\mathcal{B}}^{\tilde{I}}(X_I, Y) \end{array}$$

2. For every sequence of morphisms $I \xrightarrow{\alpha} J \xrightarrow{\beta} K$ in $\mathcal{B}\mathbf{r}$, I -labeled configuration $\tilde{I}$, J -labeled configuration $\tilde{J}$ and 2-morphisms η and ε in $\mathcal{B}\mathbf{r}$ as in the following diagram:

$$\begin{array}{ccccc} I & \xrightarrow{\tilde{\alpha}} & J & \xrightarrow{\tilde{\beta}} & K \\ & \searrow \tilde{I} & \downarrow \tilde{J} & \swarrow \tilde{K} & \\ & & \langle 1 \rangle & & \end{array} \quad \begin{array}{ccc} \Downarrow \eta & & \Downarrow \varepsilon \end{array}$$

4. Colored Braided Operads

collections of colors $X_I = \{X_i\}_{i \in I}$, $Y_J = \{Y_j\}_{j \in J}$, $W_K = \{W_k\}_{k \in K}$ and a color $Z \in \mathcal{B}$ the following diagram commutes:

make diagram prettier

$$\begin{array}{ccc}
 \prod_{j \in J} \text{Mul}_{\mathcal{B}}^{\tilde{I}_j}(X_{I_j}, Y_j) \times \prod_{k \in K} \text{Mul}_{\mathcal{B}}^{\tilde{J}_k}(Y_{J_k}, W_k) \times \text{Mul}_{\mathcal{B}}^{\tilde{K}}(W_K, Z) & & \\
 \downarrow (\gamma \circ s) \times \text{Id} & \searrow \text{Id} \times \gamma & \\
 \prod_{k \in K} \text{Mul}_{\mathcal{B}}^{\tilde{I}_k}(X_{I_k}, W_k) \times \text{Mul}_{\mathcal{B}}^{\tilde{K}}(W_K, Z) & & \prod_{j \in J} \text{Mul}_{\mathcal{B}}^{\tilde{I}_j}(X_{I_j}, Y_j) \times \text{Mul}_{\mathcal{B}}^{\tilde{J}}(Y_J, Z) \\
 & \searrow & \downarrow \\
 & & \text{Mul}_{\mathcal{B}}(X_I, Z)
 \end{array}$$

Example 4.5. Let $\mathcal{B}_{\text{color}}$ be a colored braided operad with one color X . Similar to the symmetric case, $\mathcal{B}_{\text{color}}$ induces a braided operad $\mathcal{B}$ as in Definition 3.2. For $n \in \mathbb{N}$, we set

$$\mathcal{B}(n) = \text{Mul}_{\mathcal{B}_{\text{color}}}^{b_n}(\{X\}_{1 \leq i \leq n}, X)$$

where b_n is our basepoint configuration in $E_2(n)$ as chosen in Construction 3.13. Further, we define the composition map

$$\theta: \mathcal{B}(k) \times \mathcal{B}(j_1) \times \cdots \times \mathcal{B}(j_k) \rightarrow \mathcal{B}(j)$$

where $j = j_1 + \cdots + j_k$, from the composition map γ in $\mathcal{B}_{\text{color}}$:

$$\gamma: \prod_{1 \leq i \leq k} \text{Mul}_{\mathcal{B}_{\text{color}}}^{b_{j_i}}(\{X\}^{j_i}, X) \times \text{Mul}_{\mathcal{B}_{\text{color}}}^{b_k}(\{X\}^k, X) \rightarrow \text{Mul}_{\mathcal{B}_{\text{color}}}^{b_j}(\{X\}^j, X)$$

corresponding to the unique E_1 -translation from b_j to $b_k(b_{j_1}, \dots, b_{j_k})$ as defined in Remark 3.14:

$$\begin{array}{ccc}
 \langle j \rangle & \xrightarrow{(b_{j_1}, \dots, b_{j_k})} & \langle k \rangle \\
 & \searrow b_j \quad \cong \quad \swarrow b_k & \\
 & \langle 1 \rangle &
 \end{array}$$

There is a unit

$$\eta: * \rightarrow \mathcal{B}(1)$$

which is defined by the identity morphism $\text{Id}_X \in \text{Mul}_{\mathcal{B}_{\text{color}}}^{b_1}(\{X\}, X)$. The unitality condition is satisfied as the identity morphism Id_X is the left and right unit for any composition map γ , so in particular θ . The associativity condition is also satisfied as it holds for $\mathcal{B}_{\text{color}}$: For any choice of integers

$$n, j_1, \dots, j_k, i_1, \dots, i_j$$

with

$$\sum_{1 \leq s \leq k} j_s = j, \quad \sum_{1 \leq r \leq j} i_r = i, \quad g_s = \sum_{0 \leq l \leq s} j_l \text{ and } h_s = \sum_{1 \leq q \leq j_s} i_{g_{s-1}+q}$$

the following diagram

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$$\begin{array}{ccc}
\mathcal{B}(k) \times \left(\prod_{s=1}^k \mathcal{B}(j_s) \right) \times \left(\prod_{r=1}^j \mathcal{B}(i_r) \right) & \xrightarrow{\gamma \times \text{Id}} & \mathcal{B}(j) \times \left(\prod_{r=1}^j \mathcal{B}(i_r) \right) \\
\downarrow s & & \downarrow \gamma \\
& & \mathcal{B}(i) \\
& & \uparrow \gamma \\
\mathcal{B}(k) \times \left(\prod_{s=1}^k \left(\mathcal{B}(j_s) \times \left(\prod_{q=1}^{j_s} \mathcal{B}(i_{g_{s-1}+q}) \right) \right) \right) & \xrightarrow{\text{Id} \times (\times_s \gamma)} & \mathcal{B}(k) \times \left(\prod_{s=1}^k \mathcal{B}(h_s) \right)
\end{array}$$

where s shuffles the entries of the tensor product, commutes as the composition map γ in $\mathcal{B}_{\text{color}}$ is associative for the following maps for $1 \leq s \leq k$, $1 \leq q \leq j_s$ and $h'_q = \sum_{t=1}^{g_{s-1}+1} i_t$:

$$\begin{aligned}
\alpha: \langle i \rangle^\circ &\rightarrow \langle j \rangle^\circ \\
h'_{q-1} + 1, \dots, h'_q &\mapsto g_{s-1} + q
\end{aligned}$$

with $\widetilde{\alpha_{g_{s-1}+q}} = b_{i_{g_{s-1}+q}}$ and

$$\begin{aligned}
\alpha': \langle j \rangle^\circ &\rightarrow \langle k \rangle^\circ \\
g_{s-1} + 1, \dots, g_s &\mapsto s
\end{aligned}$$

with $\widetilde{\alpha'_s} = b_{j_s}$. So we have defined a planar operad $\mathcal{B}$. It remains to see how the braid action on $\mathcal{B}$ is induced by the definition of $\mathcal{B}_{\text{color}}$. We recall from Lemma 3.8 that the fundamental group of $E_2(n)$ is the braid group up to relabeling. Let $\sigma \in B_n$ and $\bar{\sigma} \in S_n$ denoting its underlying permutation. Then we have the following commuting diagram in Br :

$$\begin{array}{ccc}
\langle n \rangle^\circ & \xrightarrow{\bar{\sigma}^{-1}} & \langle n \rangle^\circ \\
& \searrow b_n & \swarrow b_n \\
& \langle 1 \rangle^\circ &
\end{array}
\quad \begin{array}{c} \Leftarrow \sigma \Rightarrow \end{array}$$

Further, we restrict the corresponding composition map

$$\gamma: \prod_{1 \leq i \leq n} \text{Mul}_{\mathcal{B}_{\text{color}}}^{\text{Id}^{\bar{\sigma}^{-1}(j)}}(\{X\}, X) \times \text{Mul}_{\mathcal{B}_{\text{color}}}^{b_n}(\{X\}^n, X) \rightarrow \text{Mul}_{\mathcal{B}_{\text{color}}}^{b_n}(\{X\}^n, X)$$

to the collection of the identities $\{\text{Id}_X\}_{1 \leq j \leq n}$ to obtain

$$\mathcal{B}(n) \rightarrow \mathcal{B}(n).$$

This defines a braid action on $\mathcal{B}(n)$. The action is associative since it is defined via γ . Let $\sigma, \rho \in B_n$ inducing the 2-morphisms in Br :

$$\begin{array}{ccccc}
\langle n \rangle & \xrightarrow{\bar{\sigma}^{-1}} & \langle n \rangle & \xrightarrow{\bar{\rho}^{-1}} & \langle n \rangle \\
& \searrow b_n & \downarrow b_n & \swarrow b_n & \\
& & \langle 1 \rangle & &
\end{array}
\quad \begin{array}{cc} \Leftarrow \sigma & \Leftarrow \rho \end{array}$$

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For the restriction to the identities from the associativity diagram of $\mathcal{B}_{\text{color}}$ we obtain the following diagram showing $\rho.(\sigma.\mathcal{B}(n)) = (\rho \circ \sigma).\mathcal{B}(n)$:

$$\begin{array}{ccc} \mathcal{B}(n) & \xrightarrow{\rho.} & \mathcal{B}(n) \\ \downarrow \text{Id} & & \downarrow \sigma. \\ \mathcal{B}(n) & \xrightarrow{(\sigma \circ \rho).} & \mathcal{B}(n) \end{array}$$

The trivial braid acts trivially due to the unitality condition in $\mathcal{B}_{\text{color}}$. Hence, we are left to show the equivariance of the braid action regarding θ . That however follows directly from associativity diagrams in $\mathcal{B}_{\text{color}}$ for the right choice of homotopy class of paths of configurations. So, the first equivariance diagram for $\sigma \in B_n$ and $\bar{\sigma}$ denoting its underlying permutation:

$$\begin{array}{ccc} \mathcal{B}(k) \times \mathcal{B}(j_1) \times \cdots \times \mathcal{B}(j_k) & \xrightarrow{\sigma \times \bar{\sigma}^{-1}} & \mathcal{B}(k) \times \mathcal{B}(j_{\bar{\sigma}(1)}) \times \cdots \times \mathcal{B}(j_{\bar{\sigma}(k)}) \\ \theta \downarrow & & \downarrow \theta \\ \mathcal{B}(j) & \xrightarrow{\sigma(j_1 \cdots j_k)} & \mathcal{B}(j), \end{array}$$

follows for the following composition of 2-morphisms in Br :

$$\begin{array}{ccccc} \langle j \rangle & \xrightarrow{(b_{j_1}, \dots, b_{j_k})} & \langle k \rangle & \xrightarrow{\bar{\sigma}^{-1}} & \langle k \rangle \\ & \searrow \cong & \downarrow b_k & \swarrow \bar{\sigma} & \nearrow b_k \\ & & \langle 1 \rangle & & \end{array}$$

b_j (on the left diagonal), b_k (on the right diagonal)

Note, that the left triangle is defining θ and is a E_1 -translation. Furthermore, let $\rho_l \in B_{j_l}$ for $1 \leq l \leq k$ and $\rho_1 \oplus \cdots \oplus \rho_k$ denotes the sum of braidings on k blocks with each block having j_l strands. Then the second equivariance diagram:

$$\begin{array}{ccc} \mathcal{B}(k) \times \mathcal{O}(j_1) \times \cdots \times \mathcal{B}(j_k) & \xrightarrow{\text{Id} \times \rho_1 \times \cdots \times \rho_k} & \mathcal{B}(k) \times \mathcal{B}(j_1) \times \cdots \times \mathcal{B}(j_k) \\ \theta \downarrow & & \downarrow \theta \\ \mathcal{B}(j) & \xrightarrow{\rho_1 \oplus \cdots \oplus \rho_k} & \mathcal{B}(j) \end{array}$$

follows from the following composition of paths

$$\begin{array}{ccc} \langle j \rangle & \xrightarrow{\rho.b} & \langle k \rangle \\ & \searrow b & \swarrow \text{Id} \\ & \langle k \rangle & \\ & \downarrow b_k & \\ & \langle 1 \rangle & \end{array}$$

b_j (on the left curved arrow), b_k (on the right curved arrow), $\bar{\rho}$ (in the middle)

where $b = (b_{j_1}, \dots, b_{j_k})$ and $\rho = (\rho_1 \oplus \cdots \oplus \rho_k)$.

Remark 4.6. Vice versa, we can define a colored braided operad with one color $\mathcal{B}_{\text{color}}$ from a braided operad $\mathcal{B}$ in the classical sense.

if time elaborate on this

4. Colored Braided Operads

4.2 Functorial Definition of Colored Braided Operads

Next, we enrich $E_2^\otimes$ with a groupoid structure.

Definition 4.7. Let $\mathcal{B}r_*$ be the following **Grpd**-enriched category:

- (i) The objects of $\mathcal{B}r_*$ are finite sets $\langle n \rangle$ for $n \in \mathbb{N}$.
- (ii) A morphism $\alpha: \langle n \rangle \rightarrow \langle m \rangle$ in $\mathcal{B}r_*$ consists of a morphism $\alpha: \langle n \rangle \rightarrow \langle m \rangle$ in $\mathcal{F}in_*$ and a collection of configurations

$$\{\tilde{\alpha}_j \in E_2(|\alpha^{-1}(j)|)\}_{1 \leq j \leq m}$$

where each configuration $\tilde{\alpha}_j$ is labeled by $\alpha^{-1}(j)$. The composition is defined analogously to the composition map in $E_2^\otimes$ as defined above. Further, the identity morphism $\text{Id}_{\langle n \rangle}$ is given by the collection of trivial configurations $\{\text{Id}^j \in E_2(1)\}_{1 \leq j \leq n}$. The morphisms in $\mathcal{B}r_*$ form the groupoid

$$\text{Hom}_{\mathcal{B}r_*}(\langle n \rangle, \langle m \rangle) = \coprod_{\substack{\alpha: \langle n \rangle \rightarrow \langle m \rangle \\ \in \mathcal{F}in_*}} \text{Hom}_{\mathcal{B}r_*}^\alpha(\langle n \rangle, \langle m \rangle)$$

where

$$\text{Hom}_{\mathcal{B}r_*}^\alpha(\langle n \rangle, \langle m \rangle)$$

denotes the product of the groupoids of homotopy classes of paths in the configuration space $E_2(|\alpha^{-1}(j)|)$ labeled by $\alpha^{-1}(j)$ for all $1 \leq j \leq m$:

It is clear that this defines a category.

Definition 4.8. A morphism $\alpha: \langle n \rangle \rightarrow \langle m \rangle$ in $E_2^\otimes$ is called active if it is active in $\mathcal{F}in_*$, i.e. $\alpha^{-1}(*) = \{*\}$.

Remark 4.9. We can extend the definition of $\mathcal{B}r_*$ to include any finite set I with a distinguished point. However, note that that category would be equivalent to $\mathcal{B}r_*$ in the analogous way as $\mathcal{F}in_*$ is equivalent to the category of finite sets with a distinguished point.

Definition 4.10. Let $\mathcal{B}$ be a colored operad. We define a category $\mathcal{B}^\otimes$ in the following way:

- (i) The objects of $\mathcal{B}^\otimes$ are finite sequences of colors of $\mathcal{B}$.
- (ii) For $\{X_i\}_{1 \leq i \leq m}, \{Y_j\}_{1 \leq j \leq n}$ being objects in $\mathcal{B}^\otimes$, a morphism

$$\phi: \{X_i\}_{1 \leq i \leq m} \rightarrow \{Y_j\}_{1 \leq j \leq n}$$

is given by a map $\alpha: \langle m \rangle \rightarrow \langle n \rangle$ in $\mathcal{B}r_*$ together with a family of morphisms

$$\{\phi_j \in \text{Mul}_{\mathcal{B}}^{\tilde{\alpha}_j}(\{X_i\}_{i \in \alpha^{-1}(j)}, Y_j)\}_{1 \leq j \leq n}$$

in $\mathcal{B}$.

- (iii) Composition of morphisms is determined by the composition in $\mathcal{B}r_*$ and $\mathcal{B}$, i.e. for morphisms $\phi: X \rightarrow Y$ and $\phi': Y \rightarrow Z$ in $\mathcal{B}^\otimes$ with $\alpha': \langle n \rangle \rightarrow \langle m \rangle$ and $\alpha'': \langle m \rangle \rightarrow \langle k \rangle$, we define $\alpha = \alpha'' \circ \alpha'$ and

$$\{(\phi' \circ \phi)_l \in \text{Mul}_{\mathcal{B}}^{\tilde{\alpha}_l}(\{X_i\}_{i \in \alpha^{-1}(l)}, Z_l)\}_{1 \leq l \leq k}$$

where $(\phi' \circ \phi)_l$ is applying to $((\phi_i)_{i \in \alpha^{-1}(l)}, \phi'_l)$ the following composition map

$$\gamma: \prod_{j \in \alpha''^{-1}(l)} \text{Mul}_{\mathcal{B}}^{\tilde{\alpha}_j'}(\{X_i\}_{i \in \alpha'^{-1}(j)}, Y_j) \times \text{Mul}_{\mathcal{B}}^{\tilde{\alpha}_l''}(\{Y_i\}_{i \in \alpha''^{-1}(l)}, Z_l) \rightarrow \text{Mul}_{\mathcal{B}}^{\tilde{\alpha}_l}(\{X_i\}_{i \in \alpha^{-1}(l)}, Z_l).$$

Let $\mathcal{B}$ be a colored braided operad, then there is a forgetful functor

$$\pi: \mathcal{B}^\otimes \rightarrow \mathcal{B}r_*$$

In this section we will focus on characterizing π .

5. Outlook

5 Outlook

Notes on outlook:

- action operads: Could we define a op-fibration definition for any group operad?

6 Appendix

We wanted to define braided operads as functors with certain properties over a suitable category. A first intuition led us to the assumption that colored braided operads might fiber over the crossed simplicial braid group as defined in [DK]. However, we discovered that we were too quick to make this assumption. We will elaborate in the following why this did not work out as we hoped originally.

Definition 6.1. The *crossed simplicial braid group* is a category ΔB equipped with an embedding $\iota : \Delta \rightarrow \Delta B$ such that:

1. the functor ι is bijective on objects,
2. any morphism $u : \iota[m] \rightarrow \iota[n]$ in ΔB can be uniquely expressed as a composition $\iota(\phi) \circ b$ where $\phi : [m] \rightarrow [n]$ is a morphism in Δ and $b \in B_m$ is acting on $\iota[m]$ in ΔB .

Note that we sometimes refer to $\iota[m]$ by writing $[m]$ instead.

Remark 6.2. To understand composition in the category ΔB let us consider two morphisms

$$f : [m] \rightarrow [n] \quad \text{and} \quad g : [n] \rightarrow [l]$$

with the factorization $f = \phi_f \circ b_f$ and $g = \phi_g \circ b_g$. We define for every $j \in [n]$ an f -pencil p_j as a collection $[j, i_1, \dots, i_s]$ with $\phi_f^{-1}(j) = \{i_1 < \dots < i_s\}$. We call s the width of the pencil. Note that we say it is an empty pencil p_j if $\phi_f^{-1}(j) = \emptyset$.

Then we obtain a braid on m strands in the following way: First, we consider $p_1, \dots, p_n$ as n strand such that we can braid by b_g . Then, we forget the empty pencil strands; in other words we restrict the action of b_g to the subset $J \subset [n]$ such that p_j is not empty for all $j \in J$. However, for notation reason and without loss of generality, we assume that there are no empty pencils. We obtain a braiding on m strands if we consider the action of b_g on the pencils as a block braiding on m strand. Each pencil corresponds to a block having the length of the pencils width. The blocks are ordered by $[n]$.

Geometrically this can be thought of as braiding the pencils. Then, as the order within the pencil remains unchanged, we can cut a pencil $[j, j_1, \dots, j_s]$ into its s underlying strands. The resulting braid corresponds to $b_g(s_1, \dots, s_n) \circ b_f$ where s_j is the width of the pencil p_j for $j \in [n]$. Thus, we define

$$g \circ f := \phi_g \circ (b_g(s_1, \dots, s_n) \circ b_f).$$

We can define a new colored operad with the crossed simplicial braid group as its indexing sets. Again, we would like to consider the forgetful functor

$$\pi : \mathcal{O}^\otimes \rightarrow \Delta B$$

which is an op-fibration and satisfies the Segal condition. However, this causes some issues. We would assume a braided monoidal category to be an example of the new colored operad. The construction of the braided monoidal structure should be analogously to the symmetric case. Meaning, the morphism in $\sigma_0 \in \Delta B$ should induce a braiding defining a braided monoidal structure on the underlying category of $\mathcal{O}^\otimes$ using the op-fibration and Segal property. Thus, similarly to the symmetric case the following diagram would have to commute in ΔB :

6. Appendix

$$\begin{array}{ccc} [1] & \xrightarrow{\tau_0^1} & [0] \\ \sigma_0 \downarrow & \nearrow & \\ & \tau_0^1 & \\ [1] & & \end{array}$$

This is not the case in ΔB . However, we were hopeful that by fixing this we might obtain the final candidate for our op-fibration. So we define a category $\Delta B'$ from ΔB by requiring for all $n \in \mathbb{N}$ and $1 \leq i \leq n$

$$\tau_i^n \circ \sigma_i = \tau_i^n.$$

There is also a graphical interpretation of ΔB , where $\sigma_i \in B_n$ is the usual braid on n strands.

[illegible]

Order preserving maps in Δ also have graphical interpretation given by the degeneracy maps

$$\tau_i^n = \begin{array}{ccccccc} & 1 & & i & & i+1 & & n \\ & | & & | & & | & & | \\ & & & \cup & & & & \\ & & & | & & & & \\ & 1 & \dots & i & \dots & n-1 \end{array}$$

and face maps

$$\delta_i^n = \begin{array}{ccccccc} 1 & & i-1 & & i & & i+1 & & n \\ | & & | & & \downarrow & & | & & | \\ \vdots & & \vdots & & \searrow & & \vdots & & \vdots \\ 1 & & i-1 & & i & & i+1 & & n+1 \end{array}$$

So, we identify the following graphically:

A diagrammatic equation showing the cap of a loop on a vertical line. On the left, a vertical line has a loop attached to its top, which is closed by a cap. This is equal to a vertical line with a small upward-pointing semi-circular cap at its top.

As it turns out, a short graphical calculation shows that the hexagon identities hold for $\Delta B'.x$ Unfortunately, we face the next challenge as we discovered that we obtain a braid group action where we would not expect one and moreover where we not want a braid group action:

Let $\sigma \in B_n$, then for $\sigma: [n] \rightarrow [n]$ we obtain cocartesian morphisms in $\mathcal{O}_{[n]}^\otimes$ for each $X \in \mathcal{O}_{[n]}^\otimes \cong \mathcal{O}^n$, so the braid group B_n acts on $\mathcal{O}^n$. However, we would expect only to gain a symmetric group action on $\mathcal{O}^n$.

Thus, we have to conclude that the crossed simplicial braid group is not the right candidate for a functorial definition of a braided operad.

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